

**POSSIBLE SEISMO-IONOSPHERIC ANOMALIES ASSOCIATED
WITH EARTHQUAKES OCCURRED IN 2018 USING GNSS TEC: A
STATISTICAL ANALYSIS**

by

Muhammad Arqim Adil

Supervisor

Dr. Munawar Shah

Department of Space Sciences



Institute of Space Technology, Islamabad, Pakistan

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STATISTICAL ANALYSIS**

A Thesis Submitted to the
Institute of Space Technology
in partial fulfillment of the requirements
for the degree of Master of Science in
Global Navigation Satellite Systems

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Muhammad Arqim Adil

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Dr. Munawar Shah

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APPROVAL BY BOARD OF EXAMINERS

Dr. Munawar Shah

Internal Examiner

External Examiner

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Muhammad Arqim Adil

ABSTRACT

The recent advances in space-based ionosphere measurements provide more detailed information about earthquakes (EQs) precursors in the upper atmosphere and the ionosphere. In this research, we study the possible relation of EQs and ionospheric anomalies for global EQs that occurred in 2018 by analyzing Total Electron Content (TEC) from the Global Navigation Satellite System (GNSS). The selection criteria of these EQs are mainly subjected to two conditions: (1) there should be at least three International GNSS Service (IGS) permanent stations falling around the EQ preparation zone; (2) the EQ should occur during the quiet geomagnetic activity ($K_p \leq 3$, $Dst \geq -20$). We perform a statistical analysis based on sliding median and interquartile range (IQR) over TEC for 15 days before and 5 days after each EQ that shows abnormal variations within 10 days before all these EQs. Our analysis shows that the possible EQ induced TEC anomalies have deviated averagely ~ 4 TECU, with a percentage deviation of about 44%, from the corresponding upper bound (UB) during quiet geomagnetic activities ($K_p \leq 3$). Whereas the other TEC anomalies induced by moderate geomagnetic activities of ~ 4 Kp show relatively fewer deviations of about 3 TECU ($\sim 37\%$) from the corresponding UB. Additionally, it is observed that, on average, with a value of 1.2421 TECU seismically induced anomalies have deviated $\sim 5\%$ more than the geomagnetically induced anomalies. Moreover, we did not find any proportionality between TEC anomalies and magnitude of the EQ. However, the sequential EQs (including fore and aftershocks) exhibits a strong proportionality between TEC anomalies and days before the EQ, as the coming EQ exhibits TEC anomalies on almost same number of days before as its predecessor.

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LIST OF ABBREVIATIONS

GNSS	Global Navigation Satellite Systems
PVT	Position, Velocity, Time
GPS	Global Positioning System
GLONASS	Global'naya Navigatsionnaya Sputnikova Sistema
US	United States
QZSS	Quasi-Zenith Satellite System
NavIC	Navigation for Indian Constellation
MEO	Medium-Earth Orbit
FOC	Full Operational Capability
C/A	Coarse/Acquisition
BDS	BeiDou navigation satellite system
GEO	Geosynchronous Earth Orbit
IGSO	Inclined-Geosynchronous Earth Orbit
TEC	Total Electron Content
CME	Coronal Mass Ejection
SSN	Sunspot Number
HSSW	High-Speed Solar Winds
PPEF	Prompt Penetration of the Electric Fields
SAPS	Sub-auroral Polarization Stream
Ap	Planetary A-index
Kp	Planetary K-index
UT	Universal Time
AE	Auroral Electrojets
Dst	Disturbance storm-time
RI	Refractive Index
EQ	Earthquake
InSAR	Interferometric Synthetic Aperture Radar
LAIC	Lithosphere Atmosphere Ionosphere Coupling
dTEC	Detrended TEC
IGS	International GNSS Services
USGS	United States Geological Survey
ISGI	International Services for Geomagnetic Indices
SIA	Seismic Ionosphere Anomalies
STEC	Slant TEC
VTEC	Verticle TEC
IFP	Inter Frequency Bias

IQR	Inter-quartile Range
UB	Upper Bound
LB	Lower Bound
Δ TEC	Differential TEC
TEC _{obs}	Observed TEC
TEC _{UB}	Upper Bound for the calculated TEC
UTC	Universal Coordinated Time
GIA	Geomagnetically Induced Anomalies

1. INTRODUCTION

1.1 Global Navigation Satellite Systems

1.1.1 Introduction

The Global Navigation Satellite Systems (GNSS) is a standard generic term for satellite-based navigation that provides autonomous geo-spatial positioning with global coverage. Thus far, the chief purpose of the GNSS is to attain acute positioning. Since the early days, humans have always been thriving for advanced levels of positioning, navigation, and timing techniques. Determination of the targeted locations on the surface of the Earth using distinct objects has been experimented out for hundreds of years. For example, celestial navigation using stars and planets have been exploited for ages. Therefore, the recent advances in space technology have provided a great opportunity to develop high-accuracy positioning and navigation systems on a global scale [1]. The Soviet Union laid the foundations of the space era when Sputnik-I was launched in 1957. The determination of the satellite's orbit was carried out from the pattern in Doppler shifts by Sputnik's transmitted signal using a single known ground station [2]. This technique revealed new ideas to the scientific community that if a satellite position is known, the position of the radio receiver could be determined that measures the Doppler shifts in the satellite's signal. This realization proved to be the genesis of the satellite-based navigation era when Transit was launched in 1959, the first experimental navigation satellite formally known as the US Navy Navigation Satellite System [3, 4]. This system was declared publicly available in 1967 and was extensively used for precise positioning and navigation. Transit was finally decommissioned in 1996 with the dawn of the Global Positioning System (GPS) as well as Global'naya Navigatsionnaya Sputnikova Sistema (GLONASS).

Any space-borne navigation system commonly comprises three components:

- **Space Segment:** It consists of a constellation of satellites orbiting at an altitude of ~20,000 km from the center of the Earth that transmits different frequency radio signals on the microwave part of the radio spectrum.
- **Control Segment:** It consists of globally distributed (or locally) monitoring stations, a master station with a backup facility at a separate location, and ground antennas for satellite communication. The responsibility of this segment is to maintain the health of the satellite system by monitoring the broadcast ephemeris and computing and uploading the required navigation data to the satellites.
- **User Segment:** It comprises the ground-based receivers and antennas to compute position, velocity, and time (PVT).

Currently, the four GNSS and two Regional Navigation Satellite Systems (RNSS) are operating that provide high levels of accuracy and positioning. The current GNSS systems include GPS from United States (US), Russia's GLONASS, European Union's Galileo, and the Chinese BeiDou satellite navigation system. Similarly, the Japanese Quasi-Zenith Satellite System (QZSS) and Navigation for Indian Constellation (NavIC) of India are the current RNSS. The details about these systems are provided in Table 1.1.

1.1.2 Evolution

Transit technology proved to be a great success that inaugurated the future satellite navigation systems. In the late 1960s, the initial development of GPS took place that was a joint venture of the US Navy and the Air Force [2]. Eventually, the first block (Block I) of NAVSTAR GPS satellites was launched on February 22, 1978, whereas the initial operational capability of the GPS was declared on December 1993 with 24 satellites

operating in the medium-earth orbit (MEO). The Full Operational Capability (FOC) of the GPS was declared in 1995. The estimated cost for the development of GPS was about \$10 billion, while the annual operation and maintenance cost has been reported to be about \$500 million [5]. GPS provides free access to civilians on GPS L1 frequency along with coarse/acquisition (C/A) code and a navigation data message. Meanwhile, another signal is modulated on both L1 and L2, known as P(Y) code, which is accessible only to the authorized personals. New L5 and L2C signals have also been introduced for civilian use during the modernization of GPS (GPS-III) that began in 2005. Additionally, a military signal (M-Code) has been transmitted to improve jamming resistance at both L1 and L2.

Table 1.1. A brief overview of GNSS and RNSS

System	No. of Satellites	Orbits	Frequency [MHz]	Initial Service	Country	Coverage
GPS	24	MEO	L1, 1575.42 L2, 1227.60 L5, 1176.45	Dec 1993	USA	Global
GLONASS	24	MEO	L1, 1602.00 L2, 1246.00 L3, 1202.02	Sep 1993	Russia	Global
Galileo	30	MEO	E1, 1575.42 E5a, 1176.45 E5b, 1207.14 E6, 1278.75	2016-17	Europe	Global
BeiDou	35	MEO, GEO, IGSO	B1, 1561.09 B2, 1207.14 B3, 1268.52	Dec 2012	China	Global
QZSS	4	GEO, IGSO	L1, 1575.42 L2, 1227.60 L5, 1176.45 E6, 1278.75	2018	Japan	East Asia Oceania region
NavIC	7	GEO, IGSO	L5, 1176.45 S, 2492.028	2016	India	Regional

The Soviet Union launched its first GLONASS satellite on October 12, 1982. The FOC was accomplished in early 1996 with a constellation of 24 MEO satellites. This FOC accomplishment was not long-lasting as Russia's economy suffered big time after the Soviet Union fell apart. But, with the Russian Government's support, the FOC of the GLONASS was retained on December 08, 2011, and maintained subsequently [6]. The GLONASS satellites are categorized into three different categories based on their generations: the first generation are GLONASS I/II (launched in 1982), second-generation satellites are GLONASS M (in 2003), and the third-generation that launched in 2011 are GLONASS-K.

Similarly, the European Union's Galileo system was initiated with the launch of two MEO satellites launched on October 21, 2011, followed by a set of four additional satellites on October 12, 2012. By the end of 2015, six additional satellites have been launched. The expected constellation of the Galileo system will consist of 30 satellites with 24 primary and 6 spare satellites. The E1 signal of the Galileo system has the same frequency as GPS's L1. The navigation data is available publicly on both E1 and E6 frequency bands. While both the data and the pilot (data-less) signals are transmitted in the E5 frequency band [7].

Initially known as RNSS, the BeiDou navigation satellite system (BDS) launched its first three Geosynchronous Earth Orbit (GEO) satellites in 2003 and forth in 2007. The BDS system will eventually include 7 GEO, 25 MEO, and 5 Inclined-Geosynchronous Earth Orbit (IGSO) satellites. The system was declared operational publicly for China and surroundings on December 27, 2011, while FOC for this area was declared on December 27, 2012. The declaration of global coverage for the BDS system is expected in 2020. The BDS system offers two levels of service an open, and an authorized service, which will be

primarily used by the Chinese government and military protocols. Currently, the BDS system is using three frequency bands known as B1, B2, and B3, but the future BeiDou-3 system will use modernized L1/E1 and L5/E5 bands, too [8].

The QZSS is an RNSS system that will include a set of satellites in inclined orbits that will be configured in such a way that one satellite always appears near the zenith above Japan. The reason behind this configuration is the highly congested cities of Japan that are full of tall skyscrapers and GPS signals get easily blocked. The first fully operational QZSS satellite “Michibiki” was launched on September 11, 2010, which will be further joined by six more QZSS satellites to form a seven satellite constellation in the next phase [9]. An estimated budget of \$1.57 billion is expected for the development, maintenance, and deployment of the QZSS satellites [2]. Michibiki is transmitting six different signals having a signal structure that supports the compatibility with the GPS and Galileo signals and a unique L-Band Experiment, 1278.75 MHz (LEX) signal that will provide a high precision service of up to 3 cm level.

NavIC is an Indian satellite-based regional navigation system that covers an area of 50°N-30°S in latitudes and 30°E-130°E in longitudes [1]. IRNSS-1A was the first IGSO satellite in the NavIC constellation that launched on July 1, 2013. IRNSS-1B was launched on April 4, 2014, followed by the launch of a GEO satellite IRNSS-1C on October 15, 2014. Another IGSO satellite IRNSS-1D was deployed on March 28, 2015, and the constellation was finally completed after the launch of IRNSS-1E. The satellites transmit two signals in L and S bands at the frequencies of 1176.45 MHz and 2492.028 MHz, respectively. The expected real-time pseudorange-based accuracy of the NavIC is better than 20 m in primary service zones [10].

1.1.3 Applications

So far, GNSS's most common use is for positioning but its usage in everyday applications is increasing exponentially. For example, hiking and gliding are demanding high levels of navigation. Similarly, the use of high-level navigation in vehicles is increasing day-by-day. Also, the fields of marine, aviation, and defense are demanding the highest levels of accuracies. Such levels of accuracies are attained by the use of the carrier phase measurements. Fig. 1.1 highlights the major applications of the GNSS. With the help of carrier phase measurements, one can establish the accurate boundary coordinates and geodetic markers that are the demanding applications of GNSS surveying.

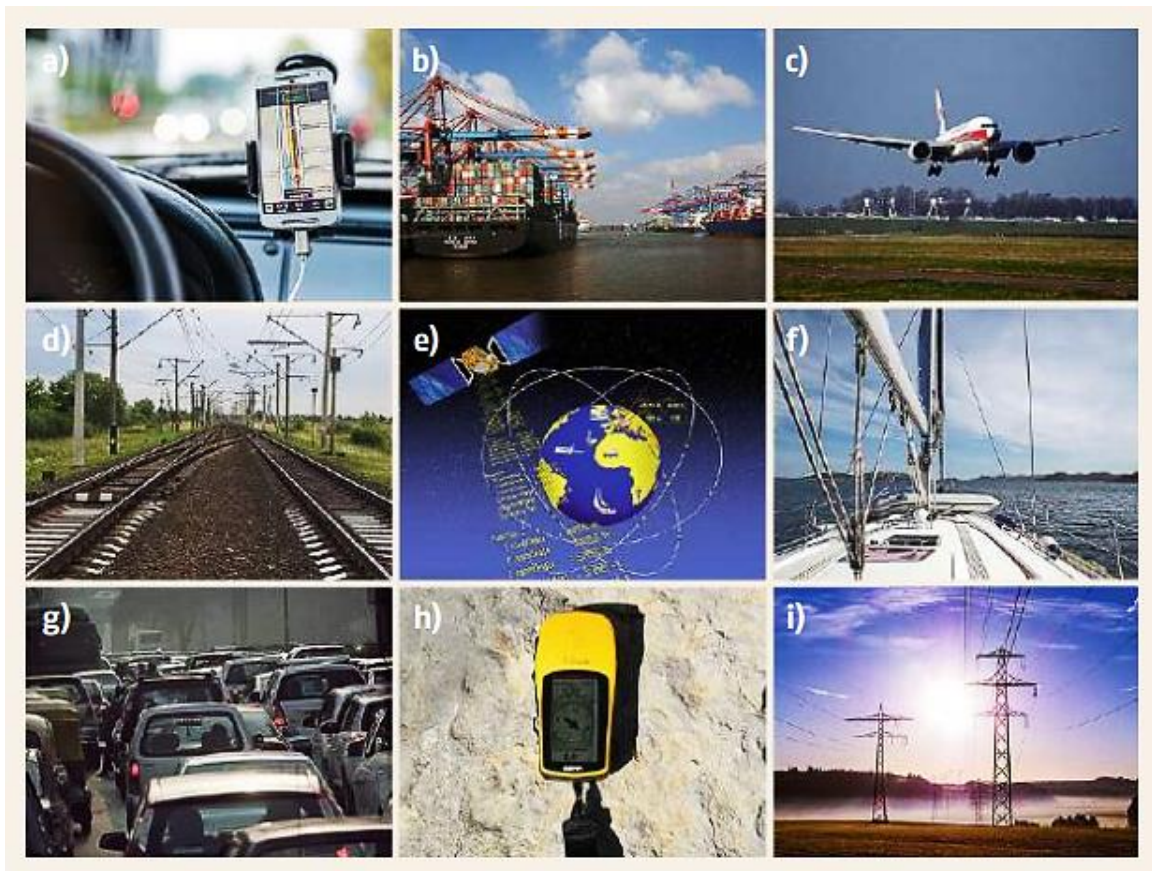


Fig. 1.1. GNSS Applications. (a.) Vehicle navigation, (b) Marine, (c) Aviation, (d) Transportation, (e) Remote Sensing, (f) Ocean navigation, (g) Traffic controls, (h) Surveying, (i) Electric Grid [1]

Nowadays, attitude control, machine control, and precision agriculture are frequently using carrier phase measurement techniques [11]. Since the GNSS has the timing accuracy as well that is used to synchronize the worldwide clocks. One of the key contributions of the GNSS timing is the synchronization of mobile phones in the telecommunication industry. Another example of GNSS timing is the alternating current phase synchronization in electrical power grids [12]. The Earth and environmental system studies using the GNSS remote sensing techniques are the new emerging applications of the GNSS. The advanced dual-frequency GNSS receivers are operating continuously around the desired geophysical locations. GNSS sounding is providing an extensive opportunity to measure the long-term tectonic plate motion [13]. Using large networks of receivers, the Earth surface activities such as land sliding and volcanos are monitored. Researchers are making significant efforts to accurately predict earthquakes by monitoring crustal motion using continuously operating GNSS networks around the seismically active regions.

Another very precious application of the GNSS is atmospheric and ionospheric remote sensing. The ionospheric and atmospheric layers impose significant delays on GNSS signal propagation, where a map of ionosphere electron content can be generated using an appropriate method on the received signals [14] and amount of water vapor in the lower atmosphere [15]. The determination of the amount of water vapors using GNSS measurements is widely being used for weather forecasting by several national weather agencies. Understanding the ionospheric conditions using GNSS measurements helps us to monitor bad weather in the ionosphere that may have severe negative effects on radio signals, navigation, and power systems. This GNSS ionosphere sensing may become an appropriate tool to develop alarming systems for earthquakes and tsunamis [16, 17].

1.2 Ionosphere and GNSS

In any GNSS system, the propagation speed of a navigation signal is critically important. If we know the propagation time of this signal from the satellite to the receiver, we can measure the ranging with the help of the product between time and speed. In the case of a vacuum, the propagation speed of an electromagnetic signal is almost equal to the speed of light. Therefore, the radio signals transmitted by the GNSS systems experience significant effects during their propagation through the Earth's atmosphere to the receiver. A dense population of free electrons, ions, and molecules resides in the region of the ionosphere that affects the propagation speed and direction of these signals. To understand the ionospheric effects on the radio signals, one should understand the inner chemistry of this layer.

1.2.1 The Ionosphere

The ionosphere is a part of the Earth's atmosphere that spreads between an altitude of ~60 km to ~1000 km. It consists of a large number of ionized gases carrying free electrons. Sun plays the primary and dominant role in the ionization of these gases. The severity of the ionospheric conditions depends upon the intensity of the solar activity [2]. There are four different layers of the ionosphere, named D, E, F1, and F2, at different altitude levels, having varying rates of production and free-electron loss. D-region is the lowest layer of the ionosphere that starts from an altitude of ~60 km and extends up to 90 km, between 90-150 km there exists E-region, and the F-region starts from an altitude of 150 km that extends to 500 km being the uppermost part of the ionosphere (Fig. 1.2).

The physical characteristics of the ionosphere largely depend upon the day/night trends. During the daytime, in the presence of the sunlight, the ultraviolet radiations from the sun

initiate a photolysis process in the ionosphere where the gas molecules (such as N_2 and O_2 at lower altitudes, and He and H_2 at higher altitudes) splits into ions and free electrons. This photochemical effect reaches the maximum during 1-2 p.m afternoon hours and starts declining afterward. During the nighttime, in the absence of sunlight, the electron-ion recombination rate takes place that ultimately reduces the free-electron count. The change in electron densities only differs by a magnitude of 1 to 2 orders between day and night.

The ionosphere also exhibits a strong coupling tendency with the thermosphere. This coupling is due to the plasma transport from the F2 layer where the plasma (trace neutral gas) has an ionization degree of about 10^{-3} [18]. These transportation processes, carried out by diffusion and neutral winds, largely depend on the thermospheric conditions. The formation of the ions and electrons, as a result of collisions between charged and neutral particles, affects neutral winds through ion drag.

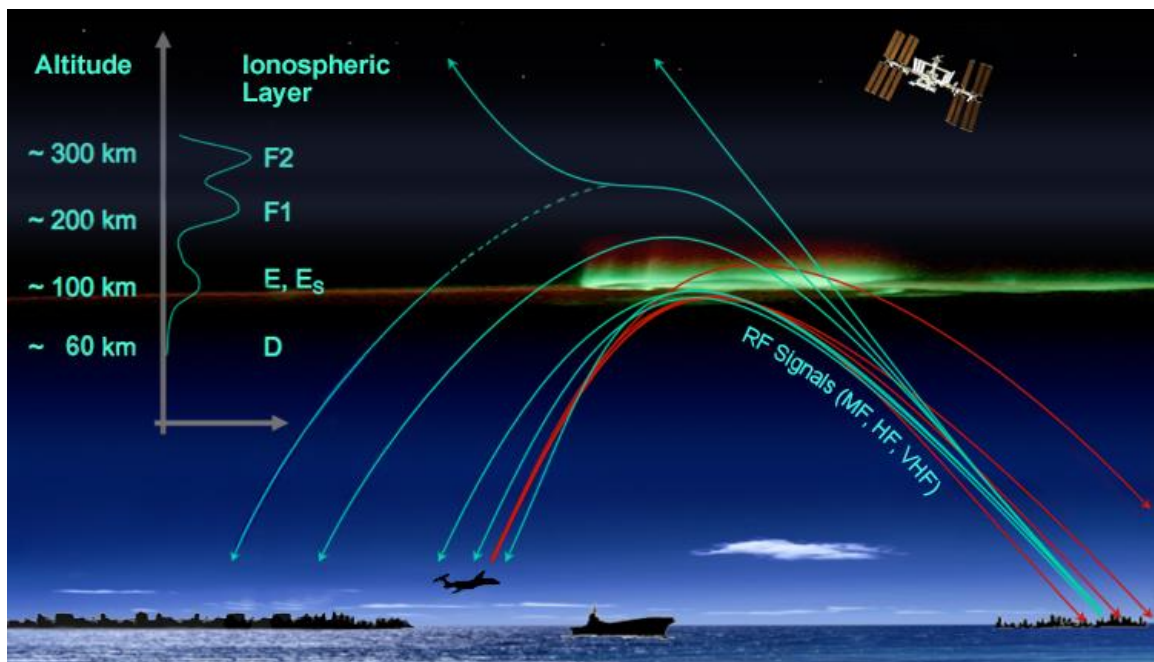


Fig. 1.2. Ionospheric layers and their effects on the radio signals [Courtesy: <http://www.astrosurf.com/>]

These neutral winds may transport the ionospheric plasma at different altitudes through the magnetic lines. On the other hand, the ions and electrons produced by the neutral wind may drive the electric fields that may induce electric currents in the ionosphere [18]. Similarly, the electric conduction from the magnetosphere may induce strong currents in the lower ionosphere. Production of these electric currents may severely influence the ionospheric dynamics by affecting the gain, loss, and transportation of the electrons.

Generally, the total electrons falling in a columnar cylinder in a unit area in the ionosphere are regarded as Total Electron Content (TEC). The total ionospheric ionization in terms of vertical TEC is largely dependent on the solar cycle. Similarly, cosmic rays and highly energetic particles originating from the solar flares may also participate in the ionosphere ionization somehow. Due to the electromagnetic radiations and highly energetic particles originating from the solar flare along with magnetosphere and thermosphere couplings, the ionosphere is very dynamic and strongly related to the space weather conditions [18, 19, 20].

1.2.2 Space Weather and The Ionosphere

Sun emits electromagnetic radiations in the form of solar wind that carries a majority of electrons, protons, and dense plasma. Intensified solar activity is characterized by enhancements in the Coronal Mass Ejections (CMEs), coronal holes, and solar flares. The solar flares are the result of abrupt energy released from the Sun that consists of bulk plasma and highly energetic particles (Fig. 1.3a). These highly energetic particles can reach the Earth's atmosphere in a fraction of an hour by easily penetrating through the ionosphere that causes significant enhancements in ionization levels, specifically, in the lower ionosphere [21]. On the other hand, the lower-energy particles traveling in the solar flare

take a few days to reach the Earth. Moreover, solar flares are usually associated with magnetic fields that are mainly attributed to sunspots. The sunspots appear as dark stains having relatively cool areas on the surface of the Sun (Fig. 1.4). The temperature of these spots may be equal to 4000 K, much lower than the surroundings. A sunspot is a result of a magnetic field disturbance during its outburst to the photosphere (Sun's visible surface) of the Sun. The formation of the sunspots may last from days to weeks, or they may persist for months [21].

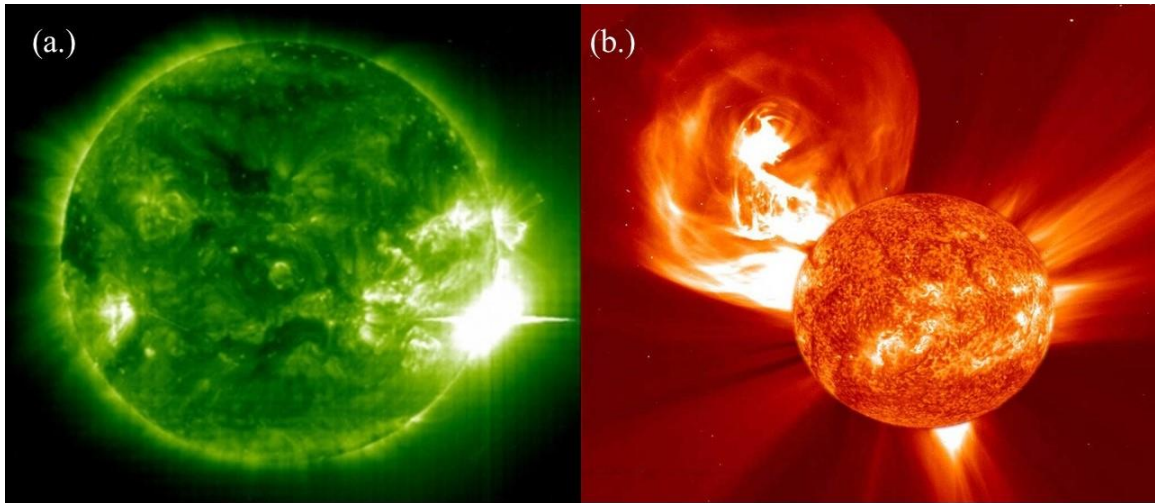


Fig. 1.3. (a.) Image of a Solar Flare's ejection from the Sun's Surface (b.) The eruption of a CME from the surface of the Sun. [Courtesy: spaceplace.nasa.gov]

The sunspots have been monitored regularly since the early 1600s, while the counting of the daily sunspot numbers (SSNs) began in 1749. The variability of the SSNs depends on the periodicity of the 11-years, known as the solar cycle. The diurnal variation of an SSN varies frequently, so long-term trends of the SSNs are computed from the moving average method. We are currently in the descending phase of the solar cycle 24 (Fig. 1.5). Around a sunspot periphery, the magnetically active regions generate CMEs in the form of gigantic bubbles carrying large plasma and highly energetic particles from the surface of the sun

(Fig. 1.3b). A single burst of a CME carries up to a billion tons of plasmas traveling at the speed of hundreds of kilometers per second. The rate of occurrence of a CME depends on the solar cycle. During the ascending phase (solar maximum), we may observe the CMEs two to three times per day, averagely. But during solar minimum, the opportunity to observe a CME reduces to only one per week. The CMEs can be extremely harmful to the space shuttles and astronauts. The majority of the CMEs coming out of the Sun misses the planet Earth except a few. The CMEs, that collide with the Earth's Magnetosphere, inject a huge amount of highly energetic particles into the upper atmosphere and lead to a global disturbance in the Earth's Magnetic field known as a geomagnetic storm [21].

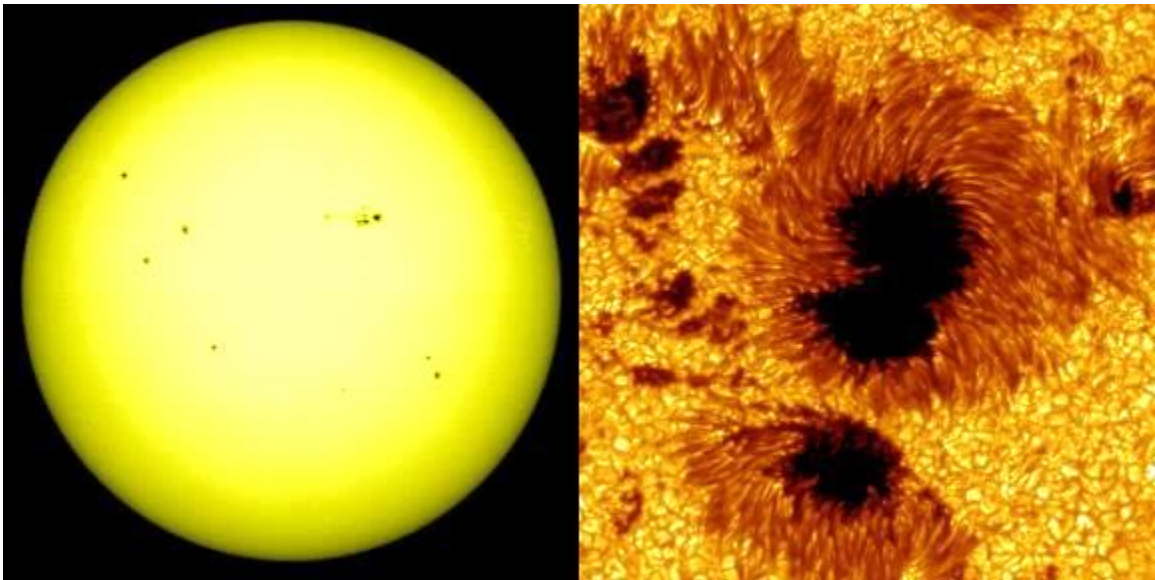


Fig. 1. 4. Multiple sunspots on the surface of the Sun (Left), and a closer look at the sunspot (Right). [Courtesy: spaceplace.nasa.gov]

The Earth is surrounded by a strong magnetic bubble (Magnetosphere) formed by the mutual interaction of the magnetic fields of both Earth and the Sun. When a high-speed solar wind (HSSW) or a CME hits the Earth, sudden bursts of the energetic particles are trapped by the Earth's magnetosphere forming tears like shape and directed towards the

ionosphere (Fig. 1.6). These particles (primarily electrons) interact with atmospheric gases such as oxygen and nitrogen molecules. The energy released from each molecule or atom, excited by the fast electrons, can be observed in the form of fluorescent colors as red or green, known as Aurora (the northern and southern lights) [21].

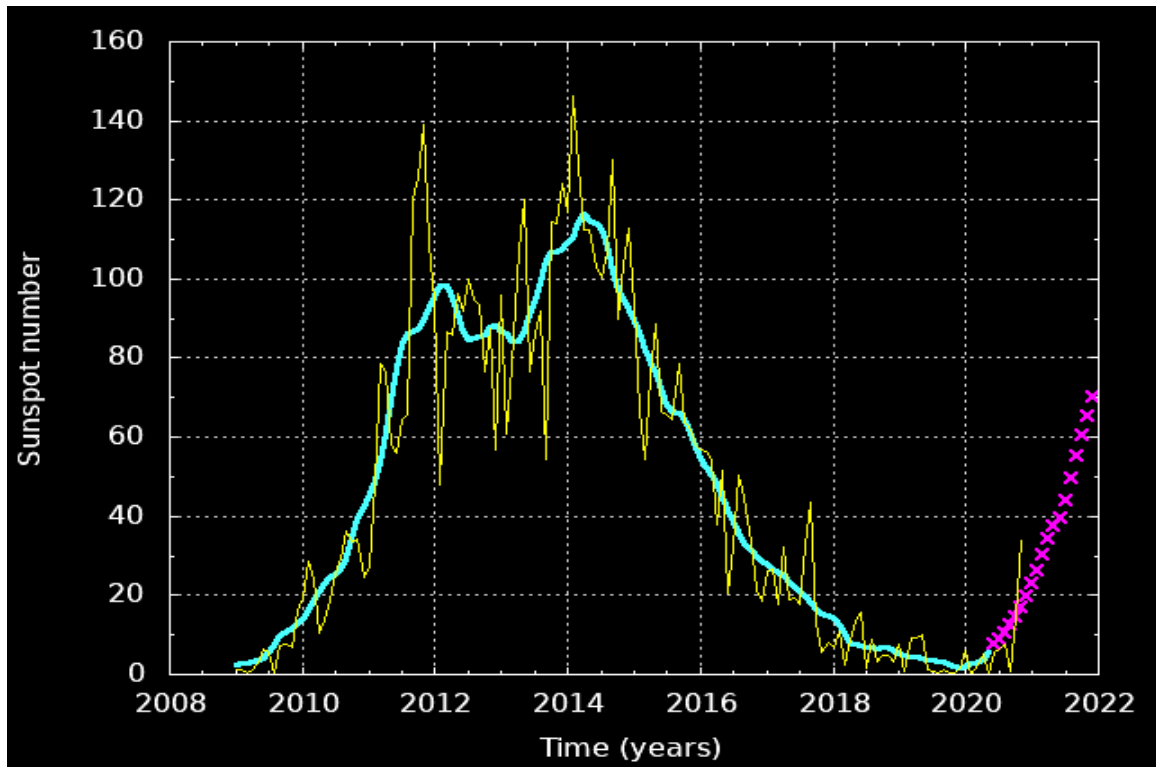


Fig. 1.5. SSNs during the solar cycle 24, between 2010-20. The year 2020 was the solar minimum during this cycle. [Courtesy: sws.bom.gov.au]

There are two separate effects associated with a large geomagnetic storm. The first effect is the formation of the equatorial anomaly (also known as Appleton anomaly) due to the prompt-penetration of the eastward electric fields (PPEFs) that are responsible for the distribution of the plasma to the equatorial ionization region [22]. The equatorial anomaly can be described as the development of the depletion in the electron density at the magnetic equator along with the increased electron density regions. Basically, the maximum enhancements in the electron density are located between ~ 15 - 20 degrees in both south and

north of geomagnetic latitudes. During a severe geomagnetic storm, the equatorial plasma is transferred to the higher altitudes by the eastward equatorial electric field using $E \times B$ forces. Consequently, the gravitational force overcomes the $E \times B$ force, at the higher altitudes, and drives down the plasma along the magnetic lines [22]. These combined effects drag the equatorial plasma more towards the north and the south poles than usual resulting in the increased electron density, at those regions, even in tranquil ionospheric conditions [22]. Furthermore, Liu and Shen, [23] identified the positive and negative phases of the geomagnetic storm over the Chinese ionosphere sector with TEC and N_e . They also reported the different mechanisms, such as ion recombination, and HSSWs induced magnetic field variations as a result of the geomagnetic storm. They suggested that the elevated plasma due to the PPEFs to higher altitudes is actually attributed to the positive storm. Conversely, during a negative geomagnetic storm, the disturbance dynamo electric fields are generated by Joule heating at higher latitudes that perambulate to the equator and may strengthen or weaken its electrojets causing a decrease in TEC [24, 25, 26].

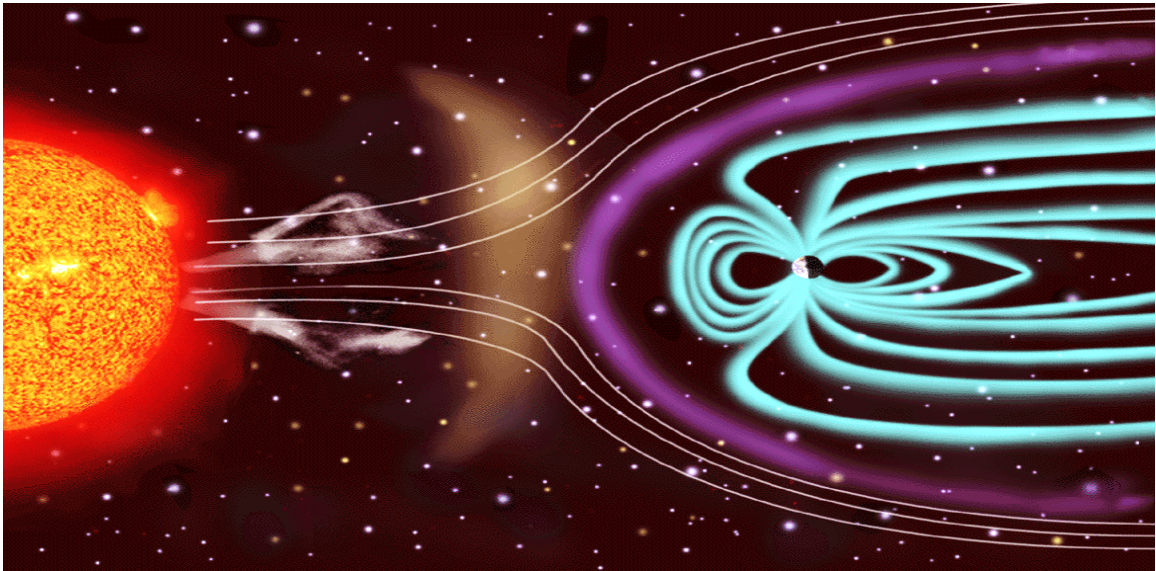


Fig. 1.6. Commencement of a geomagnetic storm. [Courtesy: hesperia.gsfc.nasa.gov]

The second effect generated by the geomagnetic storms is the evolution of the sub-auroral polarization stream (SAPS), which is a result of the complex electrodynamical coupling between the magnetosphere and the ionosphere [22]. The production of the SAPS usually takes place at the outer boundary of the plasmasphere. The direction of these SAPS electric fields is radially outward at the equator where they expand towards the poleward direction at the higher altitudes by tracing along the magnetic lines. The SAPS electric fields smear away the plasmasphere's outer boundary by overlapping the plasmasphere and reallocate the formerly equatorial plasma to the higher latitudes. Eventually, the bulk plasma of low latitudes is shifted towards the poleward and westward by $E \times B$ drift produced by SAPS electric fields.

The variations in the ionosphere caused by the magnetic storms are known as ionosphere storms. To measure the significance of the geomagnetic storms onto the ionosphere different geomagnetic indices (Ap, Kp, AE, Dst) have been introduced. For example, the impact of the geomagnetic disturbances was examined by the Ap index, and the intensity of the geomagnetic storm was considered as “strong” when the value of the planetary A-index (Ap) is 50 or more [27]. Geomagnetic activity is characterized by the magnitude relying on the values of the planetary k-index (Kp) between 0 to 9 based on the variations observed in a given 3-hour interval of the universal time (UT) [28, 29]. The Ap index has a range between 0-400 nT based on a 24-hours average of the converted Kp index values [30]. The Kp index has a very complex nature in terms of its physical interpretation because they are very sensitive to any geophysical current system, including the auroral electrojets (AE), currents in the magnetopause, and field-aligned currents. AE are horizontal electric currents formed in the polar region ionosphere. Data from auroral observatories with a

resolution of 1 minute is subtracted from the average horizontal density value in the magnetically quiescent 5 days, calculating the AE index. 1 minute of data from all observatories is sorted, resulting in the largest value (AU) and the smallest value (AL) of this data. The difference between these two values is defined as the AE index.

The formation of the increased ring currents due to ions is the principal property of a geomagnetic storm. The disturbance storm-time (Dst) index measures the decrease in the ring current component of the magnetic field on the horizontal plane in the equator. An abrupt decrease in the value of Dst shows the increased intensity of a geomagnetic storm. The Dst index is measured from near-equatorial geomagnetic observatories at 1-hour intervals [31], by making a geometric correction to the data obtained from each observatory, the measurements are normalized according to the magnetic equator. On the other hand, Kp indices are used more frequently than Dst indices for the evaluation of a geomagnetic activity because of their rapid availability within a few weeks after the observation day. Table 1.2 provides the classification of the geomagnetic storms, as mentioned by Gonzalez et al. [30].

Table 1.2. Classification of the geomagnetic indices

Geomagnetic Index	Phases				
	Quiet	Low	Moderate	Strong	Intense
Ap	≤ 20 nT	≥ 30 nT	≥ 50 nT	≥ 80 nT	≥ 100 nT
Kp	0	≤ 3	4	5-6	7-9
Dst	≥ -20 nT	≤ -30 nT	≤ -50 nT	≤ -100 nT	≤ -200 nT
AE	≤ 200 nT	≤ 400 nT	≤ 800 nT	≥ 1000 nT	≥ 2000 nT

1.2.3 Ionospheric Effects on the GNSS Signal Propagation

For any GNSS system, the satellites are the position references moving at the speed of 4 km/s in space. The distance between a satellite and a receiver is measured on the basis of signal transmission time from the satellite to the receiver. The receiver and the satellite should be perfectly synchronized in order to attain a nearly accurate transmission time. The receiver clock bias can easily affect the signal's transmission time for any satellite at any instant of the measurement. The measured ranges called pseudoranges can vary by a common amount [2]. Therefore, the receiver clock bias is the fourth unknown that needs to be sorted out along with three coordinates of the position. A geometrical view for the solution of the user position and the receiver clock bias by using pseudorange measurements and the resulting equations is depicted in Fig. 1.7.

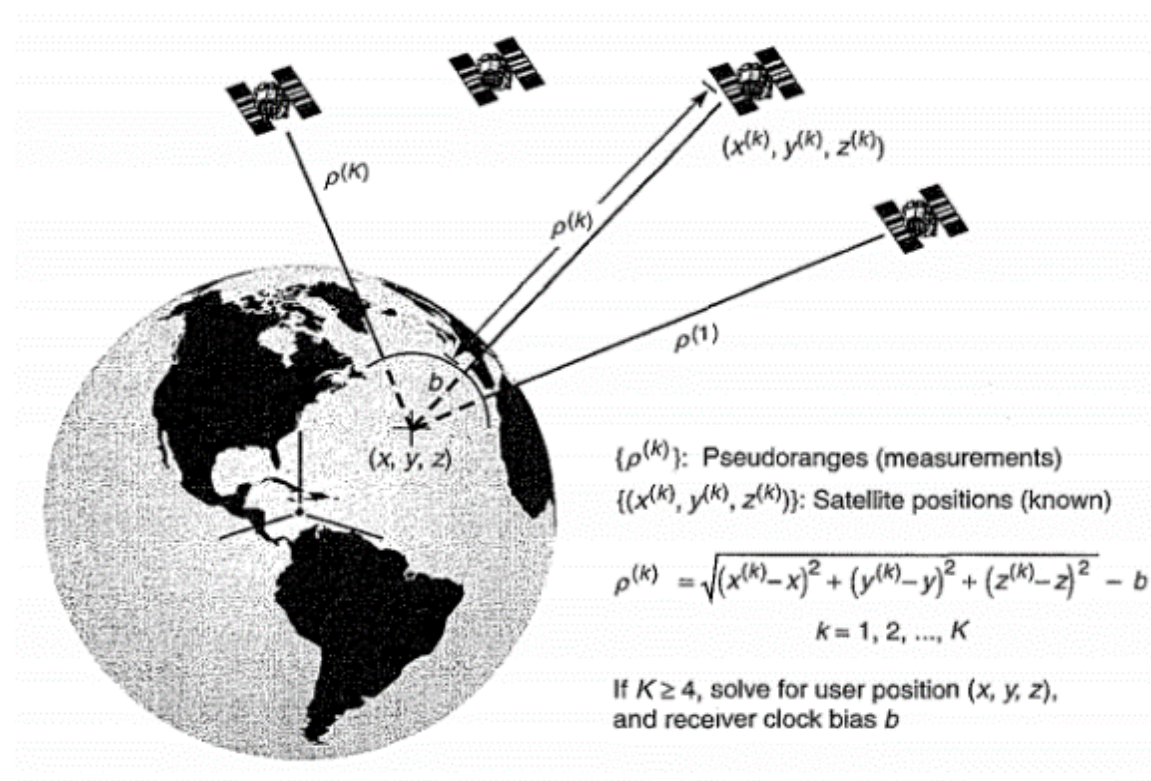


Fig. 1.7. Determination of the User Position and pseudorange equation. [2]

During their travel time from the satellite to the receiver, GNSS signals travel through the different layers of the atmosphere before reaching the ground receivers. Specifically, during their passage through the ionosphere (which is a complex part of the Earth's atmosphere dominated by the space weather phenomenon, as discussed in previous sections), they face significant delays and advances in travel time. The ionospheric effects on the GNSS performance have been widely investigated, since the advent of the radio navigation systems [32, 33]. The speed, direction, and travel time of the signals got severely affected during their passage through the ionosphere. Eventually, the carrier phase and pseudorange measurements experience severe positioning inaccuracies from 1 meter to 100 meters. These errors are measured by the integrated effect of refractive index (RI) through the signal ray path, which is the ratio of the speed of an electromagnetic signal in the vacuum with respect to air. A navigation signal is a modulated signal that carries a group of different signals of the different frequencies superposed on a single carrier frequency. In a dispersive medium, each of the group signals travels at a relatively different speed [21]. Consequently, the modulated signals will propagate at different velocities compared to the carrier, which is called group speed. The RI for the group of signals can be defined as:

$$n_g = \frac{c}{v_g} \quad (1.1)$$

$$n_g = n + f \frac{dn}{df} \quad (1.2)$$

The derivative $\frac{dn}{df}$ describes the change in the RI with respect to frequency. The changing RI will bend the path of the propagating signal through a given medium. The RI of the

ionospheric plasma is measured by the electron density, frequency of the radio wave, and the Earth's magnetic field along with several other constants including charge and the mass of the electron [21]. If we ignore the magnetic field effects, the expression for the phase refractive index, for the frequencies greater than the frequency of the ionospheric electron oscillations (~15-30 MHz), approximates to:

$$n = 1 - \frac{40.3 N_e}{f^2} \quad (1.3)$$

where f denotes the frequency in hertz and N_e represents the electron density in electrons per meter cube. Using (1.2) we can estimate the group refractive index of the ionosphere:

$$n_g = 1 - \frac{40.3 N_e}{f^2} \quad (1.4)$$

If we assume a relatively high value of N_e of 10^{12} el/m³, then n for a single GPS L1 frequency is approximately equal to 0.9999838 and the value of n_g for a modulated signal (the group of signals) is almost equal to 1.0000162 [21]. These RI values suggest that the phase speed of a single signal is much greater than the speed of light in the vacuum, while it is slightly smaller for a modulated or grouped signal. Due to the changing RI, the propagation speed of the GNSS signals varies all along their passage through the ionosphere. So, the GNSS signals received at the ground level are severely affected by the cumulated effect of the ionosphere. The propagation time for a GNSS signal to reach the earth surface is given as:

$$\tau = \int_s \frac{n'}{s} dS \quad (1.5)$$

where n' is the RI for either phase or group, and S denotes the propagation path of the signal. Now, we may calculate the range between a satellite to a receiver by multiplying the propagation time by the speed of light in the vacuum.

$$\begin{aligned}\rho_\Phi &= c\tau_\Phi \\ &= \int_S \left(1 - \frac{40.3 N_e}{f^2}\right) dS\end{aligned}\tag{1.6}$$

$$\begin{aligned}&= \rho - \frac{40.3}{f^2} \int_S N_e dS \\ \rho_p &= \rho + \frac{40.3}{f^2} \int_S N_e dS\end{aligned}\tag{1.7}$$

Here, the ρ is the true geometric range between the satellite and the receiver. From the eq. (1.7), It is obvious that the pseudorange measurements are largely dependent on the integrated electron density in the ionosphere. As a result, the GNSS signals experience a delay in the carrier measurements and an equally advanced phase. The term $\int_S N_e dS$ represents the integrated number of the electron density along the ray path of the signal also known as TEC.

As discussed in the previous sections, the space weather phenomenon dominantly controls the TEC variability. An abrupt change in the TEC values will result in a sudden change in the carrier phase, which may induce difficulties in the carrier tracking loop of the GNSS receiver. For example, a change of $0.19 \times 10^{16} \text{ m}^{-2}$ in TEC may lead to a phase change of 1 radian in the propagating signal, which may cause enough problems for the tracking loop

of a GNSS receiver [21]. These short term variations in the phase and amplitude of the signal are known as ionospheric scintillations. The scintillation effects are dominant during the evening times and may affect differently to the GNSS receivers depending on the structure of the receiver. During a sudden variation in the TEC, some code-less dual-frequency receivers may face the loss of lock problems in the L2 signal. The formation of a cycle slip is due to the temporary loss of lock and must be repaired before the processing of the navigation data. A large ionospheric TEC bias can disturb the correct determination of the number of integer cycles.

Furthermore, there exist two regions in the ionosphere of the Earth, extending between ± 20 latitudes on either side of the equatorial region to the polar regions, that can strongly cause signal fading for a shorter period of time. Sometimes, this fading becomes extremely dominant and can reduce the signal level far below the threshold of the receiver's signal lock. The signal fading occurs when an intense geomagnetic storm hits the Earth's magnetosphere and, consequently, changes the ionospheric dynamics. During this period, the navigation data are lost until the relocking of the signal. Such manipulating of the lost and found of the navigation data may persist for hours. For example, during an intense geomagnetic storm that occurred in March 1989 ($Dst = -589$ nT) in the solar cycle 22, severe TEC variations carried a change of 1 Hz in 1 second to the range-rate. Consequently, the receivers, having a bandwidth of less than 1 Hz, were losing the lock continuously during the main phase of the storm [21]. Jin and Oksavik, [34] also reported GPS scintillations during an intense geomagnetic storm that occurred on St. Patrick's Day March 17-18, 2015. They reported the losses of lock in the GPS L2 signal due to the power fading and rapid phase oscillations as a result of abrupt changes in the TEC values.

1.3 GNSS Seismology

1.3.1 Introduction

Earthquakes (EQs) may cause abrupt energy releases from the Earth's crust due to the strain effect possessed by the forces of the tectonic plate motion and mountain buildings. There are two measurable phenomena induced by the EQs: (1) the propagation of the seismic waves, radially outward, from the seismic source that causes ground shaking and can be measured by the global seismometers, (2) the crustal deformation on the Earth's surface due to the variation in the integrated elastic energy produced from the sudden slip fault on the Earth's surface [35]. The hypocenter depth of an EQ may vary between 0 to hundreds of kilometers. Hence, an EQ may take a few seconds to minutes, from the hypocenter to the Earth's surface, to induce surface ruptures. The global seismometers are playing a vital role in providing this primary shaking information [36].

However, the exponential increase in the high-rate (20 Hz) GNSS receiver networks is providing an exciting opportunity to measure and monitor the ground motion in real-time with high accuracy. The robust GNSS techniques can make it possible to measure the decisive magnitude of an impending EQ within the first 20 seconds of shaking since the initial cleavage, rather than to wait for the completion of the rupture [37]. Another apparatus, known as interferometric synthetic aperture radar (InSAR), is also very useful in monitoring crustal deformation. But the constraints of the level of coherence in the InSAR measurements arises to measure the relative displacements from the stress over wide areas as the time difference between the observations extends from weeks to months [36]. Phase coherent is an important parameter that controls the distribution of the possible interferometric measurements and spatial spread of the surface deformation, which may

come in handy in predictions of the plate motion. But the robust EQ forecasting cannot possible with InSAR imagery due to its time limits. In contrast, the third generation GNSS signals will provide improved accuracy and enhanced capability in site velocities in near future, with continuously operating permanent GNSS stations will provide high rate measurements of the seismic ruptures. The combined use of the InSAR and GNSS measurements may play a crucial role in EQ forecasting. The mutual GNSS and InSAR measurements over fault boundaries will monitor and identify the areas where a strain is building up relatively slow within a country. Such diversity can be found in the New Zealand country where both kind of regions, with a high strain rate and low strain rate, exists. In New Zealand, slowly deforming regions still have a strong tendency to produce a large EQ [38]. So, the mutual observations of the GNSS and InSAR may play a significant role in monitoring the crustal deformation and, hence, predicting an EQ.

1.3.2 Ionospheric Monitoring for Earthquake Forecasting Using GNSS

As discussed in the previous section, GNSS is a useful tool for monitoring crustal deformation, but it can also be very helpful in monitoring the seismic ionospheric precursors. The first detection of the ionospheric precursors was reported by Davies and Baker, [39] during their analysis of the M 9.2 Alaska EQ that occurred on March 28, 1964, by utilizing the doppler records and vertical ranging ionograms. Similarly, the formation of the acoustic gravity wave due to the crustal deformation during the preparation period of this EQ was also detected [40]. These preliminary findings suggested that the abrupt energy, released by the fault rupture, not only propagates upwards to the Earth's surface but also propagates further up in the atmosphere and changes the ionospheric dynamics. The seismically induced ionospheric variations occur as a result of the strong amplification

of the gravity waves propagating upward to the ionosphere. This amplification is the result of the mutual effect of the exponential reduction of the atmospheric density and the conservation of kinetic energy [41]. Thus GNSS satellites, which are orbiting about 20000 km away from Earth, may provide an extensive opportunity to monitor these ionospheric disturbances (Fig. 1.8). Therefore, ionospheric TEC from ground and satellite measurements provided significant evidence to monitor the ionospheric conditions over the epicenter of future EQs during the quiet geomagnetic activity [42, 43, 44, 45]. For example, Liu et al. [46] examined significant decreases in TEC anomalies within 5 days before $20 \geq 6.0$ magnitude EQs in the Taiwanese regions during 1999-2002, by using statistical analysis based on the running median-quartile methods. Similarly, Kon et al. [47] reported intensified TEC anomalies within 1-5 days before $M_w \geq 6.0$ EQs in Japan during a long term of 12 years (May 1998-May 2010) by using the ground-based GNSS receivers that were located within 1000 km from the corresponding epicenters. Moreover, Liu et al. [48] reported negative GPS TEC anomalies associated with large magnitude EQs in China during 1998–2008 and provided a substantial statistical correlation of lithosphere and ionospheric anomalies. For Japanese EQs, Shah et al. [49] also reported combined anomalous patterns in the form of positive TEC and critical frequency of F2 (f_oF2) layer perturbations within three days before the mainshock which were the results of the anomalous lower atmosphere specifying the propagation path of the ionospheric anomalies. Pikridas et al. [50] examined the TEC variations prior to the M 6.7 Aegean Sea EQ and found a significant decrease in TEC values 1-day before the main event. They further suggested that these anomalies exhibit the location of the epicenter as well as the lithosphere-ionosphere phenomenon.

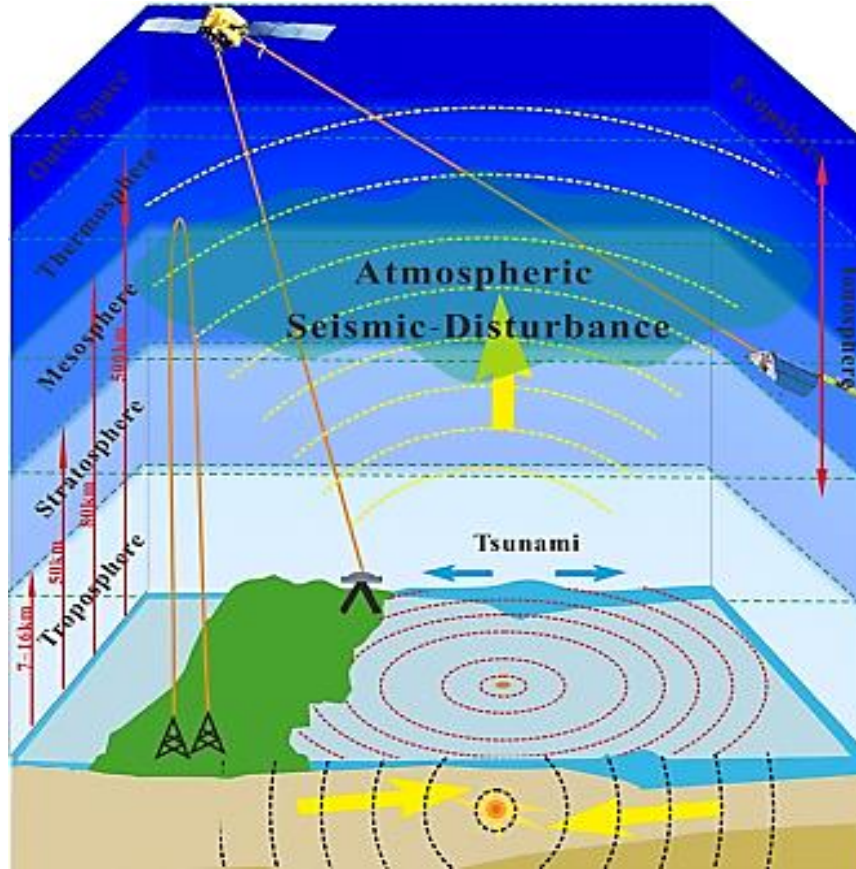


Fig. 1.8. The ionospheric detection of seismic precursors using GNSS sounding [41].

On the other hand, some studies strongly suggest that the ionosphere variations are dominantly controlled by the geomagnetic storms (as described in the previous sections) and even denied the existence of EQ induced ionosphere anomalies with the use of TEC [51, 52] and explained that EQ forecasting is not possible with the current cluster of ionosphere monitoring satellites. For example, Afraimovich et al. [53] reported that the TEC anomalies recorded before October 16, 1999, Hector Mine EQ were largely contributed by geomagnetic activity, rather than seismically induced ionosphere perturbations. Similarly, Dautermann et al. [54] monitored the ionospheric TEC variations during the preparation periods of two EQs with $M > 6$ using a large network of 256 continuously operating GPS receivers in the Los Angeles basin. The overall averaged TEC

time-series were observed to be highly correlated with the day-to-day variability and space weather phenomenon, however, no conclusive evidence was found regarding the seismic ionospheric precursors. They further suggest that: there should be some proper mechanism that can explain the evolution of the seismic precursors in the ionosphere.

1.4 Focal Mechanisms Defining Seismic Lithosphere-Ionosphere Coupling

It is difficult to directly relate the ionospheric variations with an EQ activity especially considering the altitudinal differences that extend from zero to more than hundreds of kilometers. It is true that the EQs are the source of abrupt energy release, as discussed in previous sections, but how this energy propagates and influence the ionosphere?, Can this energy directly penetrate the ionosphere?, If yes, then what are the factors that are involved in the transportation of this energy?. On the other hand, if this energy is not reaching the ionosphere directly, then what are other sources that establish the lithosphere and ionosphere interplay?. These are the concerns that should properly be addressed. According to Freund [55], the energy released during the tectonic plate motion carries electromagnetic emissions over a wide frequency range. The emissions consist of the highly mobile charges named positive holes that travel rapidly towards the surface of the Earth during inter-tectonic stress. As more and more positive holes gather at the surface of the Earth, when their cumulative number density exceeds a threshold value, they initiate the air ionization and ions recombination with a strong local electric field at the lower atmosphere. This air ionization leads to variate the ionospheric electron density that affects the radio wave propagation through the ionosphere (Fig. 1.9). In another analysis, Daneshvar and Freund, [56] further explained the propagation of these positive ions in the atmosphere. They suggested that the rise in the positive airborne ions through the atmosphere is primarily

attributed to the thermal up-drift caused by the latent heat energy produced by the moisture condensation and vertical penetration of the global electric field. This upwelling electric field then broke up into cells, at the upper atmosphere, and further proceeds towards the ionosphere, ultimately causing ionospheric plasma variations.

Pulinets et al. [57] proposed an Earth's Lithosphere-Atmosphere-Ionosphere Coupling (LAIC) phenomenon by explaining the synergy of several processes that may occur during a seismic event. Their proposed mechanism is largely based upon the emanation of the radon gas that plays a key role in the initialization of the several chemical and physical processes in the atmosphere and, consequently, disturbs the ionospheric electron densities.

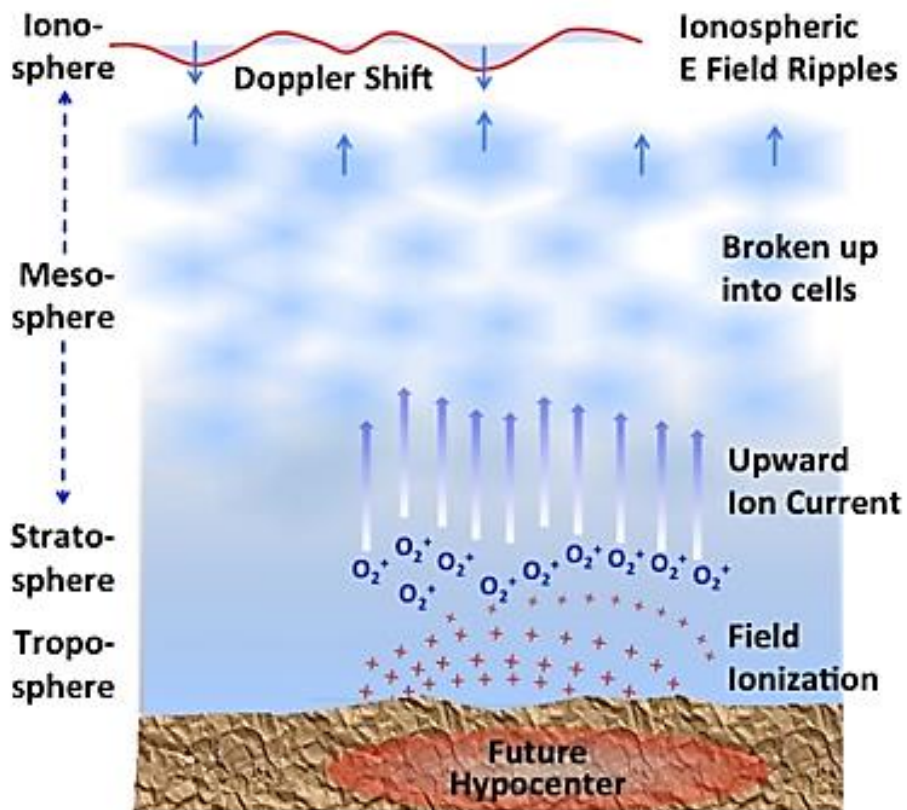


Fig. 1.9. The formation of the positive holes due to the stress effect of the igneous rocks and their interaction with the ionosphere during a tectonic activity [56].

They suggested that enhanced radon secretions are produced by stressed effects of rocks inside Earth's crust during tectonic activity. These radon increments initiate several physical and chemical processes at the Earth's surface where other gases such as methane, CO₂, Hydrogen, and Helium serve as carriers. These radon increments produce ion nucleation by ionizing the lower atmospheric air. The production of air ionization is a result of the condensation of the vapor vapors that further releases the latent heat of vaporization to the upper atmosphere. Meanwhile, a significant reduction in the mid-tropospheric relative humidity occurs as a result of this air ionization. This reduction in the relative humidity amplifies the condensation process and enhances the production of the latent heat, which further produces a heat spot at the top of the atmosphere called outgoing longwave radiations. The formation of the large ion clusters due to these atmospheric circulations changes the atmospheric boundary layer conductivity. The currents produced by this effect further varies the ionospheric plasma concentrations (Fig. 1.10).

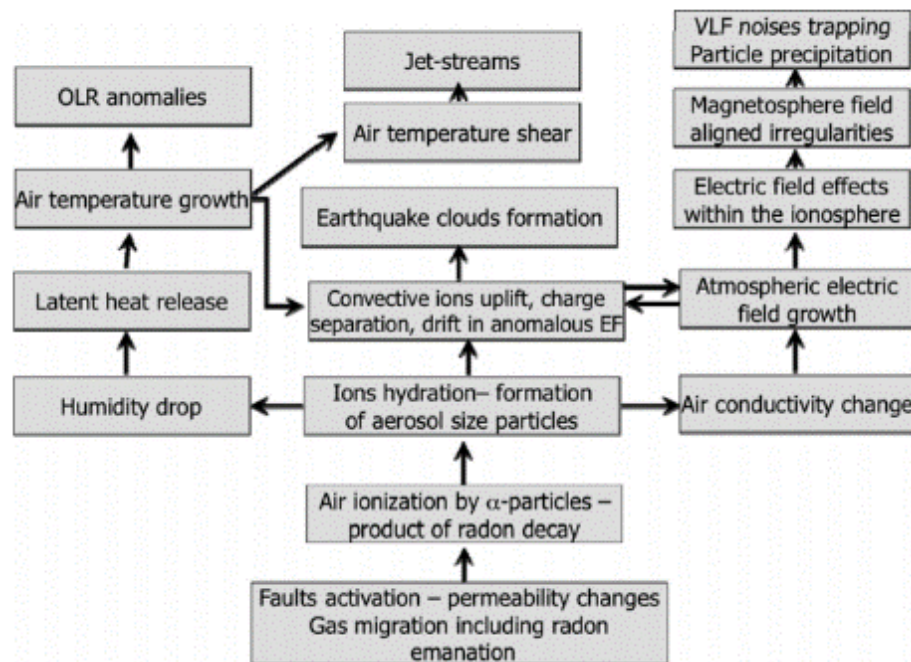


Fig. 1.10. The LAIC phenomenon proposed by Pulinets et al. [57]

Furthermore, the production of these charge carriers at the atmospheric boundary layer is accompanied by the ascension of charge carriers over the surface of the Earth, which produces the vertical electric field. This electric field uplifts vertical current flow in the atmosphere, which further leads to enforcing upward-downward flows in negative and positive ions, respectively, in the ionosphere, over the EQ preparation zone. These current dynamos can drive plasma bubbles which might lead to TEC variations and may cause 1-20% changes in detrended-TEC (dTEC) [58].

Explaining the atmospheric currents in the LAIC phenomenon, Sorokin and Ruzhin [59] stated that: during the geological processes (such as an EQ), the intensive processes initiated at the lithosphere and lower atmosphere have an electrodynamical impact on the ionospheric electron cloud. They further explained the involvement of the quasi-static electric field between Earth's surface and the ionosphere, during the seismic activity, as a vertical structure amplifying the electrical conduction along the altitudes through the ionospheric layers. Their calculations showed that: the quasi-static electric field may attain a value of ~ 10 mV/m in the ionosphere, whereas its value at the ground level is ~ 100 mV/m.

1.5 Motivation and Objectives

Since GNSS signals provide a deep insight over ionospheric conditions by estimating the amount of TEC content in the ionosphere. The researchers have proposed that ionospheric TEC concentrations may also be influenced by the seismic activity which may help in understanding the EQ-ionosphere interactions. But it has been discussed earlier that the decisive role is played by the solar and geomagnetic activities in changing the ionospheric dynamics. So, it becomes very difficult to associate an ionospheric variation with the EQ during active solar or geomagnetic activity. In this research, I will investigate the possible

relationship between the EQs and the ionospheric variations for the EQs of $M_w > 5$ that occurred in 2018 using GNSS-TEC. The TEC datasets are obtained from the permanent GNSS stations operating around the corresponding EQ's premises. 2018 was an active year of solar and geomagnetic disturbances, hence finding an EQ ionospheric precursor will be challenging. To avoid this difficulty, I have designed an EQ selection criterion based on two conditions that will help us to monitor the geomagnetically-free ionosphere environment. The selection criteria will be: (1) there should be at least three International GNSS Service (IGS) permanent stations falling around the EQ preparation zone; (2) the EQ should occur during the quiet geomagnetic activity ($K_p \leq 3$, $Dst \geq -20$). The quantification of the ionospheric TEC anomalies will be performed based on statistical analysis. All the analysis, techniques, and figures will be performed under the MATLAB[®] 2019a environment.

2. DATA AND METHODOLOGY

2.1 Case Studies

In this thesis, the ionospheric conditions are investigated before global EQs of $M_w > 5.0$ occurred in 2018 across the world. The EQs information was retrieved via (<https://earthquake.usgs.gov/earthquakes/search/>) from the United States Geological Survey (USGS). The total number of $M_w > 5.0$ EQs, which occurred in 2018, is about 2000. So, to explore the possible seismically induced ionospheric perturbations during 2018, we defined selection criteria based on two conditions. The first condition is: the selected EQ should occur in a quiet geomagnetic activity period ($K_p \leq 3$, $Dst \geq -20$). Since the geomagnetic activities are the major drivers of ionospheric variabilities, so, a period with a quiet geomagnetic activity will provide a sturdy inside into the induction of seismic ionospheric anomalies (SIAs). We find that the greatest number of EQ preparation periods were falling inside the abrupt geomagnetic activities and hence, excluded. The second condition is: there should be at least three IGS stations falling around the EQ preparation zone. The EQ preparation zone is calculated by Dobrovolsky et al. [60] formula:

$$R = 10^{0.43M} \text{ km} \quad (2.1)$$

where M is the magnitude of the EQ and R denotes the radius of the critical region in km. So, the remaining EQs, which were occurred during quiet geomagnetic conditions, were further filtered according to the second condition. Finally, after a meticulous analysis, we choose 11 EQs that fulfill the defined conditions. The detailed information about these selected EQs and the critical region is provided in Table 2.1. Furthermore, the geographical locations of these case studies are illustrated in Fig. 2.1.

Table 2.1. The detailed information about the selected EQs.

Event	Region	Mw	Date	Location of the Epicenter		Depth	EQ Preparation Zone
				Latitude	Longitude		
1	Alaska	5.6	21-11-2018	59.955°N	153.266°W	143.3	255.86
2	Alaska	7.1	30-11-2018	61.346°N	149.955°W	46.7	1129.80
3	Argentina	5.4	28-01-2018	31.273°S	68.668°W	86.9	209.89
4	Argentina	5.2	01-02-2018	31.562°S	70.040°W	100.7	155.96
5	Chile	5.6	08-02-2018	37.439°S	73.979°W	4.3	255.86
6	Greece	6.8	25-10-2018	37.520°N	20.557°E	14	839.46
7	Honduras	7.5	10-01-2018	17.483°N	83.520°W	19	1678.80
8	Russia	7.3	20-12-2018	55.100°N	164.699°E	16.6	1373.21
9	Russia	6.2	25-12-2018	55.542°N	166.450°E	11.2	463.45
10	Taiwan	6.1	04-02-2018	24.157°N	121.708°E	12	419.76
11	Taiwan	6.4	06-02-2018	24.134°N	121.659°E	17	564.94

2.2 Geomagnetic Indices Data

To monitor the geomagnetic influences in the ionosphere during these seismic events, we obtained the geomagnetic indices datasets from the International Services of Geomagnetic Indices (ISGI; <http://isgi.unistra.fr/>). The datasets of the geomagnetic Dst index with a temporal resolution of 1 hour, whereas the datasets of the geomagnetic Kp index with a temporal resolution of 3 hours were obtained for each analyzed EQ. The ionosphere perturbations are largely controlled by the AE index rather than the Kp or Dst index at higher latitudes, but its data was not available during 2018, till now.

2.3 GNSS-TEC Data

To obtain the TEC time series for the monitoring of the ionospheric conditions during the 11 EQs, we choose three IGS stations according to the defined criteria. For the two Alaska EQs, we have used two nearby stations (FAIR, GCGO) from Fairbanks city, Alaska, and one (WHIT) from Whitehorse city of Canada (Fig. 2.1a). Similarly, two stations (SANT, ANTC) from Santiago and Los Angeles, Chile, and the other one (CORD) from Cordoba, Argentina have been utilized for the construction of the TEC time-series for the three EQs that occurred in South America. Macedonia's ORID, Bulgaria's SOFI, and MATE station from Italy are used for the ionospheric investigation of the M 6.8 Greece EQ. Furthermore, SSIA, MANA, and SCUB stations from the surroundings of the Caribbean Sea are used to obtain TEC values for the large EQ of M 7.5 that occurred offshore Honduras.

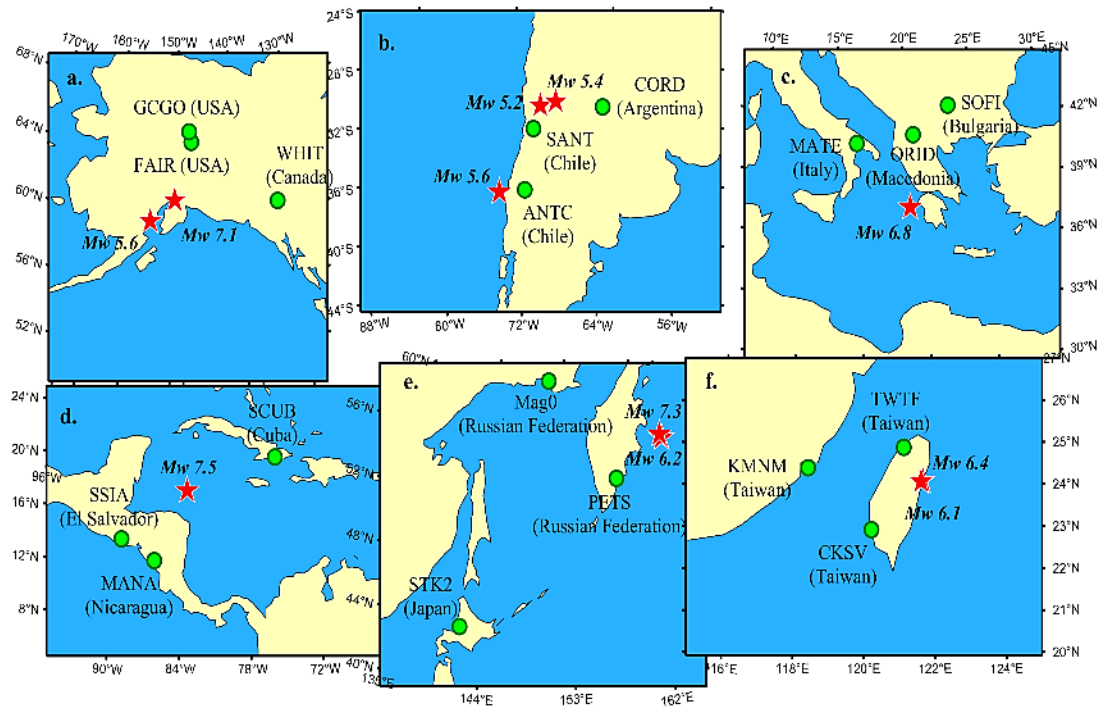


Fig. 2.1. The geographical locations of 11 EQs along with IGS GNSS stations. The red stars represent the epicenter, whereas green dots denote the IGS stations

The ionospheric conditions of two EQs that hit the Russian region have been monitored by estimating the TEC values from two stations in Russia (PETS, MAG0) and one from Japanese territory Shintosukawa (STK2). Lastly, for Taiwanese EQs three stations, operating within Taiwan, are used named TWFT, CSKV, and KMMN.

2.4 Estimation of the Ionospheric TEC

To construct a TEC time series, the GNSS signals recorded at each IGS station (mentioned above) were retrieved via (<ftp://cddis.gsfc.nasa.gov/pub/gps/products/ionex/>) at a sampling rate of 30 s using the IONOLAB software which is publicly available from the online interface (<http://www.ionolab.org/>). The STEC (Slant TEC) through the line of sight l between the satellite and receiver can be calculated as [61]:

$$I = \frac{40.3}{f^2} \int_s N_e ds = \frac{40.3}{f^2} STEC \quad (2.2)$$

Where I denotes the ionospheric delay for signal L having frequency f . To compute the carrier phase measurements and linear combinations of the pseudo-ranges (geometry free) because the STEC cannot be calculated directly from eq (2.2) due to the lack of direct measurement of geometric range [62, 63].

$$P = P_1 - P_2 = I_1 - I_2 + b_s + b_r + \varepsilon_p \quad (2.3)$$

In the above equation P_1 , P_2 , b_r , and b_s are the pseudo-ranges and differential code biases for receiver and satellite, respectively, whereas ε_p denotes the noise effects on pseudo-range. Similarly, for the carrier phase measurements, the equation is given as:

$$\Phi = \Phi_1 - \Phi_2 = I_2 - I_1 + \lambda_1 N_1 - \lambda_2 N_2 + B_s + B_r + \varepsilon_l \quad (2.4)$$

where Φ_1, Φ_2, B_r , and B_s represents carrier phase measurement and Inter Frequency Biases (IFBs) with λ and N are wavelength for signal L and integer ambiguity of phase measurement, respectively, whereas ε_l represents the multipath error. Now combining both Eqs. (2.3) and (2.4) [64]:

$$P + \Phi = \lambda_1 N_1 - \lambda_2 N_2 + B_s + B_r + \varepsilon_l + b_s + b_r + \varepsilon_p \quad (2.5)$$

Here, we can neglect the term ε_l for the carrier phase measurements as its value is negligible as compared to ε_p for pseudo-range measurement. Now, as IFBs are considered as constants, so the term $(P + \Phi)_{arc}$ in an incessant arc could be measured as an average value as [65]:

$$\begin{aligned} (P + \Phi)_{arc} &= \frac{1}{n} \sum_{i=1}^n (P + \Phi)_i \\ &= (\lambda_1 N_1 - \lambda_2 N_2)_{arc} + B_s + B_r + b_s + b_r + (\varepsilon_p)_{arc} \end{aligned} \quad (2.6)$$

Here, n is the total number of measurements for a single arc. Further, a subtraction of eq (4) from eq (6) can terminate the ambiguity terms as well.

$$\tilde{P} = (P + \Phi)_{arc} - \Phi \approx I_1 - I_2 + b_s + b_r + (\varepsilon_p)_{arc} - \varepsilon_l \quad (2.7)$$

$\tilde{P}$ is the smoothed pseudo-range ionospheric observation. Now, STEC can be obtained by substituting the eq (2.2) into eq (2.7) [33].

$$STEC = \frac{f_2^2 f_1^2}{40.3(f_2^2 - f_1^2)} \left(\tilde{P} - b_s - b_r - (\varepsilon_p)_{arc} + \varepsilon_l \right) \quad (2.8)$$

Finally, the STEC can be converted to vertical TEC (VTEC) [62]:

$$VTEC = STEC \left(\cos \left(\sin^{-1} \left(\frac{R \sin Z}{R + H} \right) \right) \right) \quad (2.9)$$

Where R is the radius of the Earth, H is the height of the thin shell (in this study, H= 350 km for the Klobuchar model) and Z is the zenith angle of the satellite.

2.5 Statistical Method for the Anomaly Detection

To identify the ionospheric anomalies associated with 11 EQs, a total of 18 dual-frequency GNSS receiver stations within the respective epicentral zones are used in the analysis. From all the GNSS stations, daily TEC with the 30-second resolution has been analyzed statistically with the confidence bounds method. The confidence bounds are obtained by the median ($\tilde{X}$) and associated Inter Quartile Range (IQR). For example, the confidence bounds of the analysis day are calculated from the median and IQR of 10 days before and 10 days after the observed day. Then, the upper bound (UB) and lower bound (LB) are as below.

$$UB = \tilde{X} + (1.5 \times IQR) \quad (2.10)$$

$$LB = \tilde{X} - (1.5 \times IQR) \quad (2.11)$$

Equations (2.10-11) show the implementation of upper and lower bound on normal TEC distribution. Moreover, TEC values beyond confident bounds manifest abnormality

associated with EQs. The recorded EQ anomaly occurs statistically within the range of 95% confidence interval.

Another method, differential TEC (ΔTEC), has been utilized to quantify EQ induced ionosphere anomalies. We calculate the difference between observed TEC and UB for detecting positive EQ induced TEC enhancements. The observed TEC (TEC_{obs}) is subtracted from UB to get the deviation in TEC from the normal distribution, as below:

$$\Delta TEC = \begin{cases} TEC_{obs} - TEC_{UB} & \text{if } TEC_{obs} > TEC_{UB} \\ 0 & \text{if } \Delta TEC < 0 \end{cases} \quad (2.12)$$

Moreover, deviation from normal TEC distribution for every GNSS station is estimated in percentage deviation. After implementing these methods, we can easily discriminate anomalous TEC between possible seismic activity and any other geomagnetic activity.

$$\% \Delta TEC = \begin{cases} \frac{TEC_{obs} - TEC_{UB}}{TEC_{UB}} \times 100 & \text{if } TEC_{obs} > TEC_{UB} \\ 0 & \text{if } \% \Delta TEC < 0 \end{cases} \quad (2.13)$$

The above equation provides a positive percentage of ΔTEC ($\% \Delta TEC$) values deviating from the corresponding UB.

3. RESULTS AND DISCUSSION

3.1 Ionosphere TEC Behavior to the Two Sequential Alaskan EQs

One of the two EQs occurred 65 km South-West of Redoubt Volcano, Alaska at 18:21:44 UTC (Universal Coordinated Time) with a M_w of 5.6 and having 143.3 km depth, on November 21, 2018, the epicenter was located at (59.955°N, 153.266°W) in latitude and longitude, respectively as shown in Table 2.1. Moreover, this EQ was followed by a gigantic EQ of M_w 7.1 that occurred on November 30, 2018, at 17:29:29 (UTC) about 14 km North-West of Anchorage, Alaska. This EQ was occurred at 47 km depth, located at (61.346°N, 149.955°W) in latitude and longitude, respectively, as shown in Table 2.1. Fig. 2.1a represents the geographical locations of the epicenters of these two EQs along with the locations of the 3 IGS stations used for TEC measurements to monitor the ionospheric conditions prior to these EQs. Figure 3.1 presents the geomagnetic activity and the TEC variations along with associated UB and LB for an observational period between Nov 10 – Dec 5, 2018. Fig. 3.1a-b shows the geomagnetic activities during the EQs observational period where a threshold value of 3 is set for the Kp index represented by a red dashed line. Moreover, if a value exceeds the threshold line, it will be considered an active geomagnetic activity. Fig. 3.1b indicates that Kp index values overlap the threshold on December 03, 2018 (3 days after 7.1 M_w EQ), declaring an active geomagnetic activity. From Fig 3.1c-e, clear TEC variations can be observed, for event 5.6 M_w , the TEC variations about 2-3 days before the mainshock can be observed at all stations, simultaneously. Similarly, a TEC deviation, about 3 days before 7.1 M_w EQ, can also be seen at both IGS stations combinedly.

On the other hand, Fig. 3.2 describes the differential TEC with respect to UB and LB. The figure describes a clearer view of the discrimination of the TEC variations between SIAs and geomagnetic influences. In the case of 5.6 M_w EQ, all IGS stations FAIR, WHIT, and GCGO record a maximum ΔTEC values of about 1.9661, 1.4762, and 1.4103 about 2 days before the impending EQ, respectively, as shown in Fig. 3a-b. A percentage TEC deviation of about 60.40%, 58.87%, and 54.31 from the UB, respectively, is recorded at all stations, as described in Table 3.1. Similarly, in the case of 7.1 M_w EQ, a maximum ΔTEC values of about 1.5995, 1.1516, and 1.2682 with a %TEC deviation of about 56.71%, 26.7%, and 73.2% was recorded, at FAIR, WHIT, and GCGO stations, 3 days before the EQ, respectively, as shown in Fig. 3.2a-c and Table 3.1.

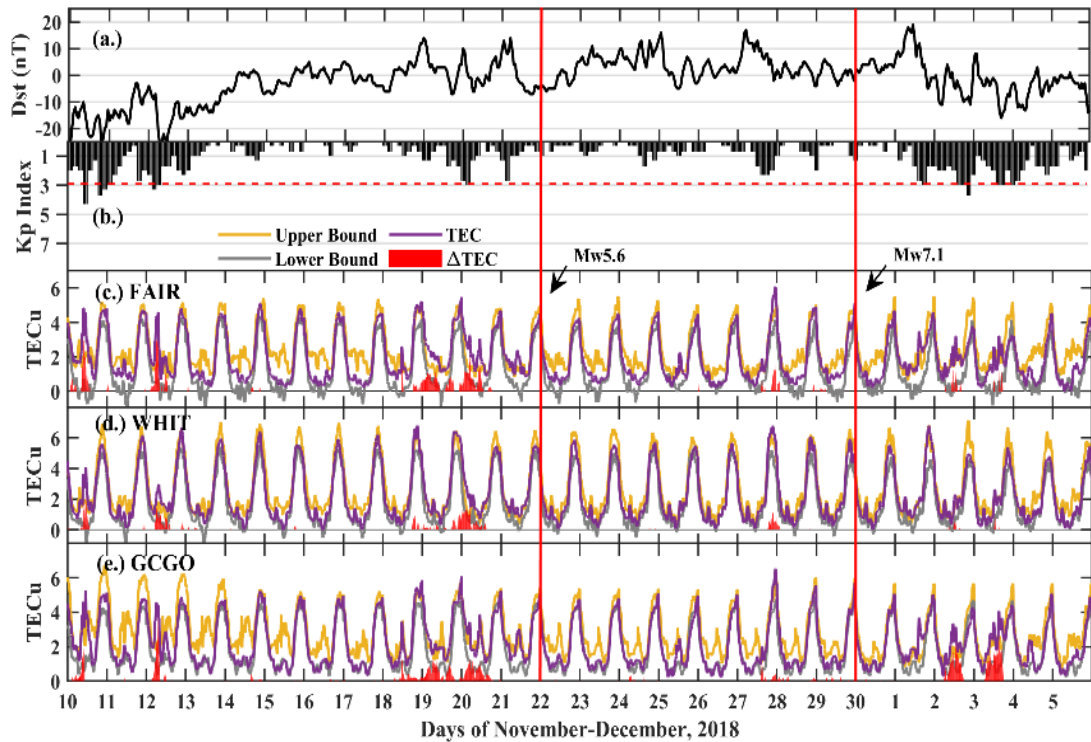


Fig. 3.1. The geomagnetic activity represented by (a.) Dst and (b.) Kp Index in the context of sequential Alaskan EQs during 2018. GNSS TEC from FAIR, WHIT, and GCGO stations are also shown in lower panels

On the other hand, Fig. 3.2 also gives a clear demonstration about TEC anomalies that whether they are associated either with a geomagnetic activity or with possible seismic activity. Several geomagnetic activities can be observed during the analysis where the value of Kp index is recorded as (4-), whereas at the other days of November 18, 19, 20 (3, 2, and 1 day before the 5.6 M_w EQ) and 27, 2018 (3 days before 7.1 M_w) the Kp indices are recorded less than 3. However, only TEC variations observed on November 18, 19, 20, and 27 are considered as the possible SIAs as shown in Fig. 3.2a-c. The figure also highlights an interesting fact that both sequential EQs have shown a strong proportion in terms of anomalies exhibition between anomalies and number of days before their corresponding shocks as depicted by Fig. 3.1c-e and Fig. 3a-c.

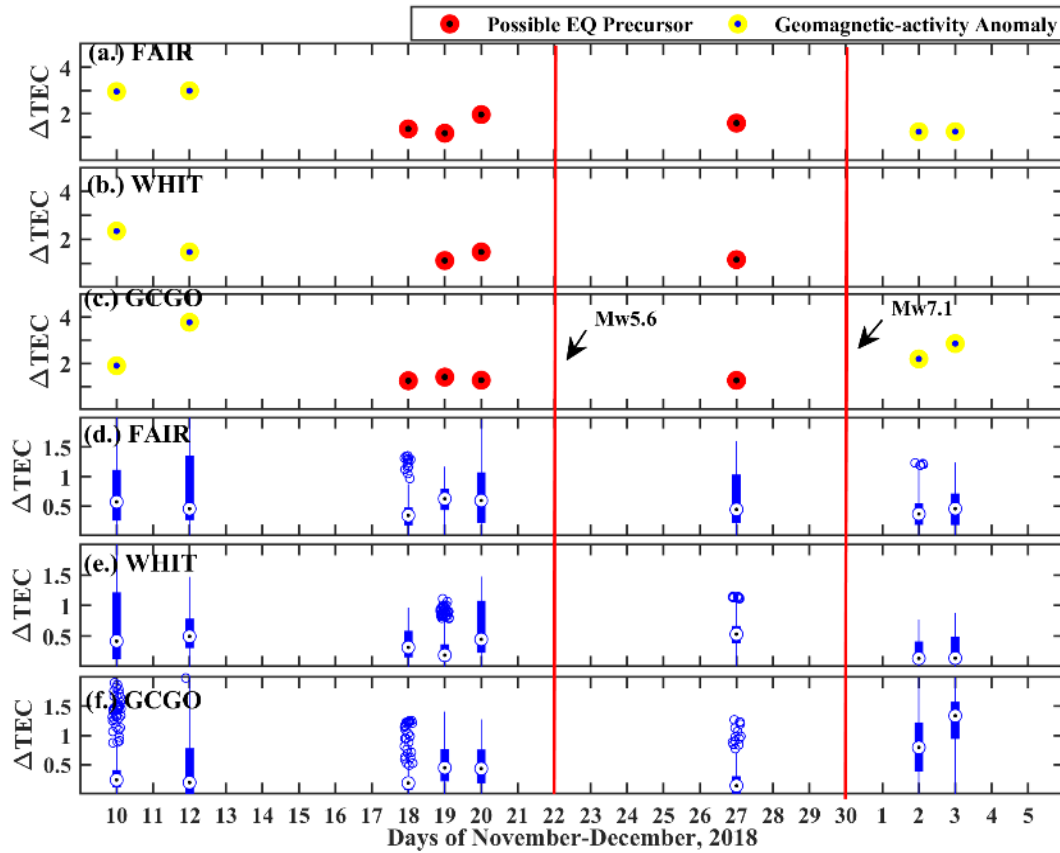


Fig. 3.2. The maximum ΔTEC and percentage deviation of TEC for Alaska EQs

3.2 TEC Anomalies Before Three EQs Occurred in South American Region

During January and February 2018, three sequential EQs hit Argentina and Chile region. An EQ of 5.4 M_w occurred 22 km north-west of Albardon, Argentina at 23:53:14 (UTC) on January 28, 2018, having a depth of 86.9 km, the epicenter was located at (31.273°S, 68.668°W) in latitude and longitude, respectively. This EQ was then followed by an aftershock on February 01, 2018, at 23:13:54 (UTC) having a M_w of 5.2 occurred 65 km south-west of Calingasta, Argentina, having a depth of 100.7 km. Furthermore, these EQs were tailed by another EQ on February 08, 2018, that occurred at 35 km north-west of Lebu, Chile at 21:19:24 (UTC) having a hypocentral depth of 4.3 km. The details of these EQs are given in Table 2.1 and their geographical locations are represented in Fig. 2.1. TEC variations accompanied by associated UB and LB along with geomagnetic activity over an observational period of Jan 16-Feb 13, 2018, is shown by Fig. 3.3. These figures show prominent variations 3 days before each EQ is analyzed in this region. Fig. 3.4a-b-c rigorously distinguishes seismic and geomagnetically induced anomalies (GIAs). Before M_w 5.4 EQ, each IGS station named as ANTC, CORD, and SANT, shows a deviation of about 5.7777, 7.2442, and 6.66 TECU respectively, with a percentage deviation of about 43.01%, 20.82%, and 28.04% as shown in Table 3.1. Similar %TEC deviation before M_w 5.2, which was an aftershock to M_w 5.4 EQ, was recorded as 6.6847, 5.9479, and 3.6215 TECU with a %TEC deviation about 48.66%, 29.09%, and 31.47% respectively. Furthermore, in the case of Chile EQ that occurred 6 days after following these EQs, a deviation of about 7.1465, 6.3939, and 4.9666 were recorded, on the stations, as listed above, with a %TEC deviation about 37.49%, 25.01%, and 36.90%, respectively.

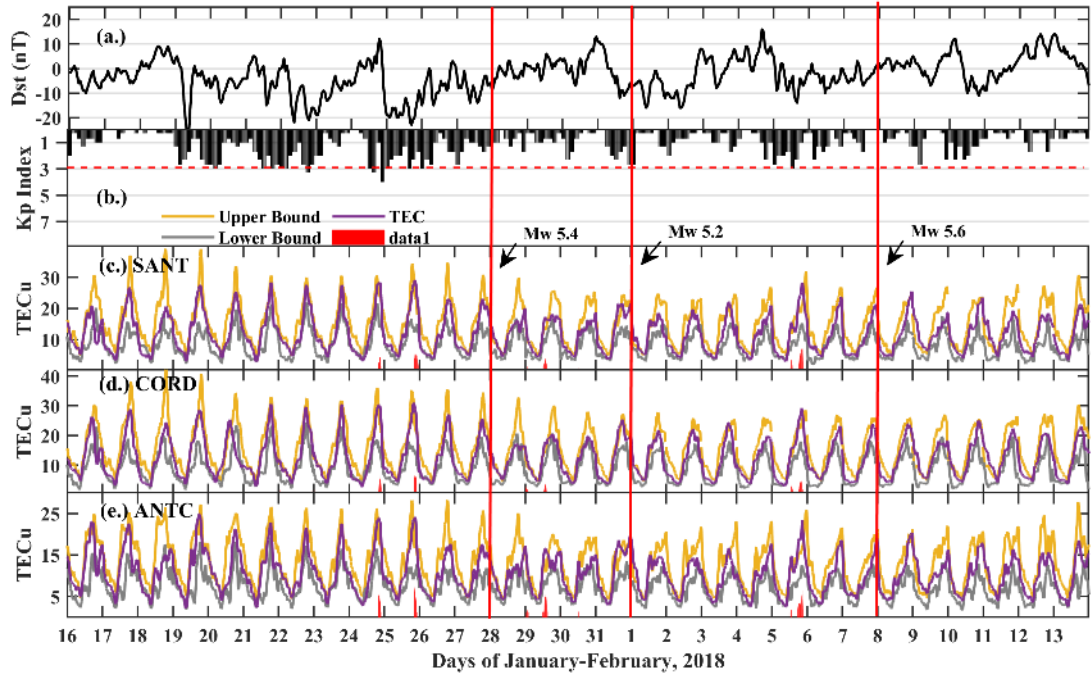


Fig. 3.3. (a.) Dst Index, (b.) Kp Index. The panels (c, d, e) present the TEC time-series for SANT, CORD, and ANTC stations. Solid red lines denote the EQ day.

Fig. 3.4 demonstrates a broader view to separate seismic and stormy anomalies. A geomagnetic storm was detected on January 24, 2018, during the period of these sequential EQs of Argentina and Chile where the Kp index was at 4 as shown in Fig. 3.3b, which caused disturbances in TEC measurements as shown in Fig. 3.4a-b-c. Whereas the other days, where TEC anomalies were found are, January 25, 2018 (3 days before M_w 5.4 EQ), January 29, 2018 (3 days before M_w 5.2 EQ), and February 5, 2018 (3 days before M_w 5.6 EQ), respectively, as shown in Fig. 3.4, where the geomagnetic activity is relatively quiet as the value of Kp remains less than 3. Table 3.1 provides complete detail about TEC deviations caused by the seismic effect during this period. Furthermore, these sequential EQs are exhibiting a healthy proportion to each other as all found anomalous 3 days before the upcoming EQs.

Table 3.1. Maximum Δ TEC values recorded at IGS stations used for Alaska, Argentina, and Chile EQs

Event	IGS Station Name	Location of the IGS Station		Δ TEC value	%TEC deviation	Region
M _w		Latitude	Longitude	TECU		
5.6	FAIR	64.978°N	147.499°W	1.9661	60.40 %	USA
	WHIT	60.750°N	135.222°W	1.4726	58.87 %	Canada
	GCGO	65.639°N	147.843°W	1.4103	54.31 %	USA
7.1	FAIR	64.978°N	147.499°W	1.5995	56.70 %	USA
	WHIT	60.750°N	135.222°W	1.1516	26.71 %	Canada
	GCGO	65.639°N	147.843°W	1.2682	73.20 %	USA
5.4	SANT	33.150°S	70.669°W	6.66	28.04 %	Chile
	ANTC	37.339°S	71.532°W	5.7777	43.58 %	Chile
	CORD	31.528°S	64.470°W	7.2442	29.01 %	Argentina
5.2	SANT	33.150°S	70.669°W	3.6215	31.47 %	Chile
	ANTC	37.339°S	71.532°W	6.6847	48.66 %	Chile
	CORD	31.528°S	64.470°W	5.9479	29.09 %	Argentina
5.6	SANT	33.150°S	70.669°W	4.9666	36.90 %	Chile
	ANTC	37.339°S	71.532°W	7.1465	37.49 %	Chile
	CORD	31.528°S	64.470°W	6.3939	25.01 %	Argentina

3.3 Ionospheric TEC Response to M_w 6.8 Greece EQ

A large EQ of M_w 6.8 occurred at 22:54:52 (UTC) in 32 km south-west of Mouzaki, Greece on 25 October 2018. The epicenter was located at (37.520°N, 20.557°E) in latitude and longitude, respectively, with a depth of 16 km. The results of geomagnetic activity, TEC, and Δ TEC variations at MATE, ORID, and SOFI stations over an observational period

between October 10-20, 2018, are shown in Fig. 3.5 and 3.6, respectively. From Fig. 3.5, This EQ induced significant ionospheric perturbations at three GNSS stations (MATE, ORID, and SOFI). The consolidated image of geomagnetic activity and TEC from three stations for October 10-30, 2018 showed abnormal ionospheric anomalies 4 days before the mainshock (Fig. 3.6). The variations of TEC and ΔTEC manifest positive anomalies on October 21, 2018 (4 days before the mainshock), consequent to abnormal energy emanation during the EQ preparation period (Table 3.2). Meanwhile, the geomagnetic activities are observed closely during this Greece event. The threshold value for the Kp index is 3, which is represented by a dashed red line (Fig 3.5b).

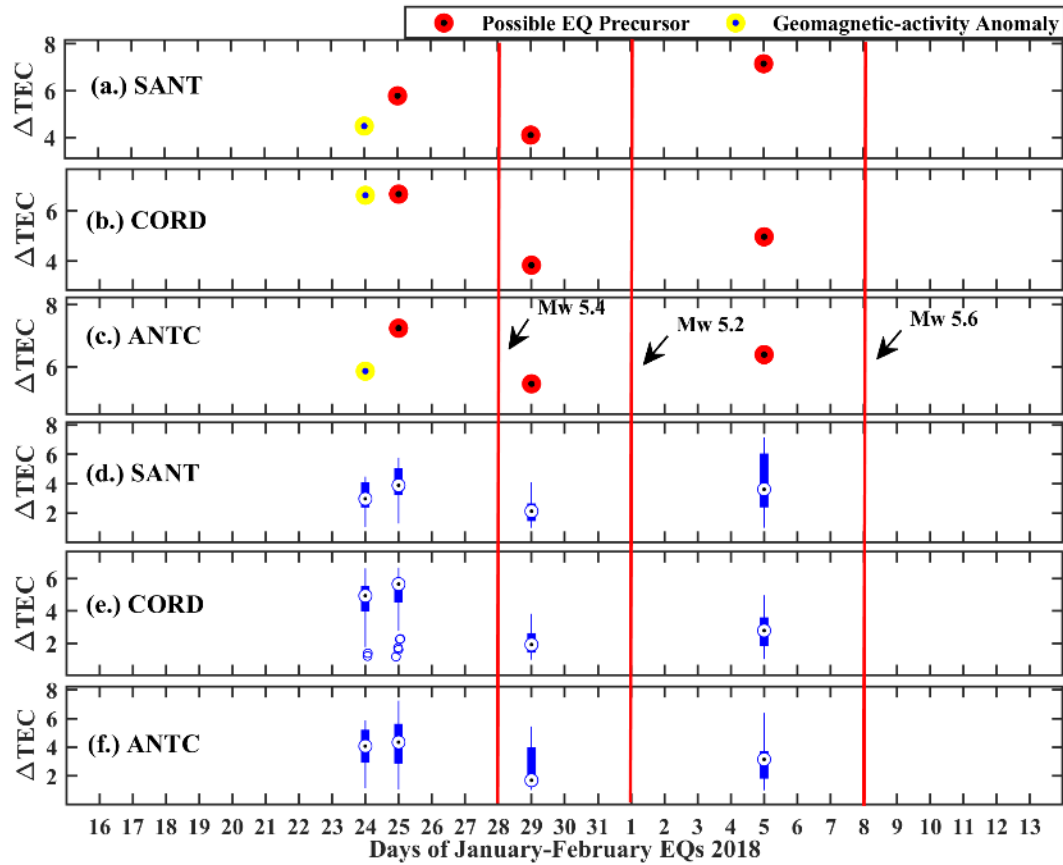


Fig. 3.4. The maximum ΔTEC differentiating the seismic and storm variations recorded at three IGS stations named SANT, CORD, and ANTC for Argentina and Chile EQs.

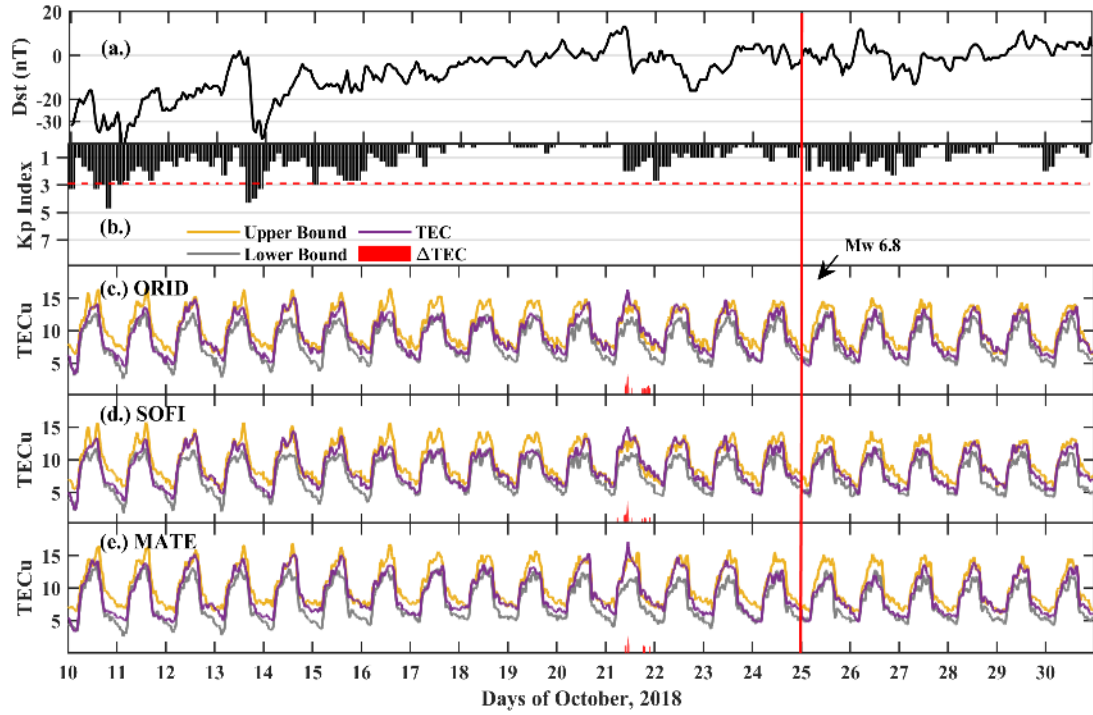


Fig. 3.5. Geomagnetic Dst and Kp Index during Greece EQ event where the red dashed line represents the threshold of quiet storm. GNSS-TEC measurements from ORID, SOFI, and MATE stations with bounds and legend are shown. The solid red line represents the EQ day.

So, the anomalies observed on October 21, 2018, are considered as possible pre-EQ seismic anomalies, as they found in quiet geomagnetic conditions ($Kp < 3$) as shown in Fig. 2a-b. The deviation in TEC on abnormal days related to EQ is recorded as 2.7546, 3.4679, and 3.7315 TECU, at MATE, ORID, and SOFI stations, respectively, which corresponds to 19.23%, 27.61%, and 32.90% in percentage deviation (Table 3.2). Similar findings are reported by other scientists. For example, Liu et al. [46] investigated a decrease in TEC anomalies within 5 days before $20 \geq 6.0$ EQs in the Taiwan region between September 1999-December 2002. The geomagnetic activities are observed closely during this Greece event. Table 3.2 provides a brief detail about the TEC anomalies observed during this event.

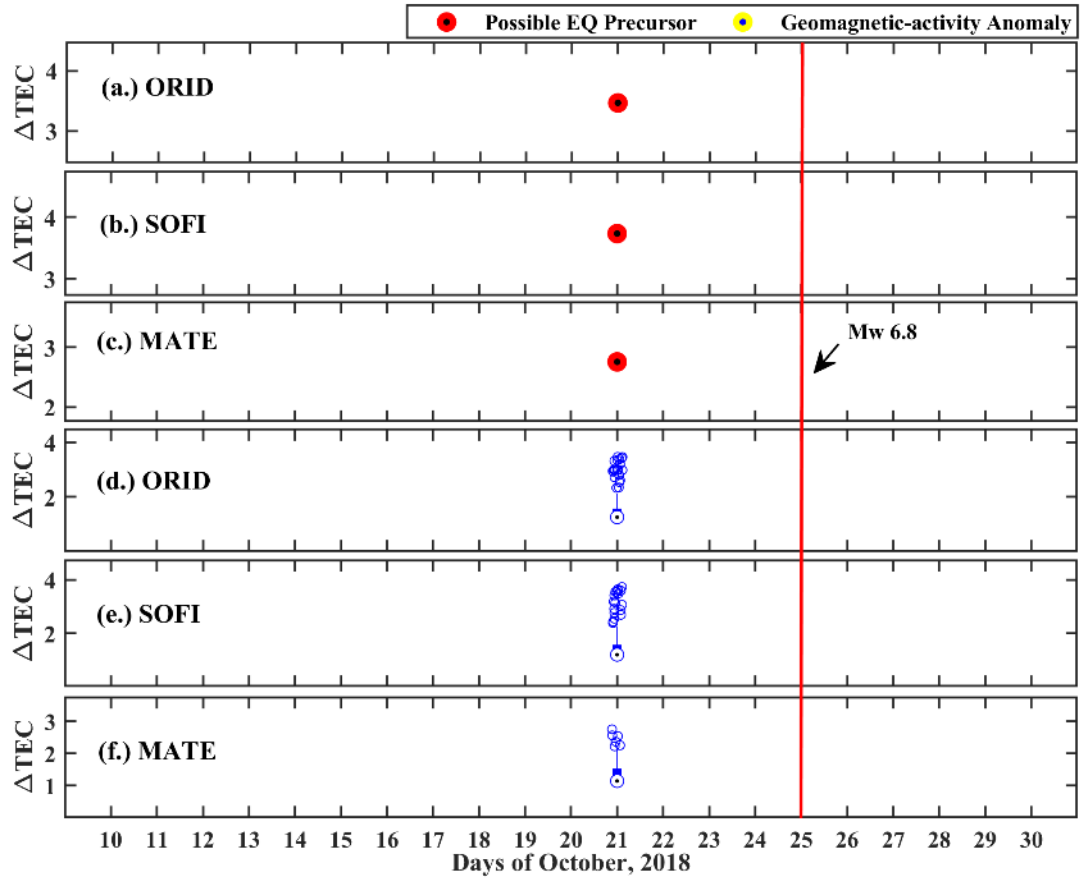


Fig. 3.6. The upper panels (a. b. c.) show maximum ΔTEC deviations associated with Greece EQ. Similarly, the lower panels show percentage deviations.

3.4 Ionospheric TEC Variabilities During M_w 7.5 Honduras EQ

The EQ of M_w 7.5 occurred at 44 km east of Great Swan Island, Honduras, at 02;51:33 (UTC) on January 10, 2018. This event manifested large ionospheric TEC enhancements that were recorded at three IGS stations MANA, SCUB, and SSIA falling within its preparation zone. Fig. 3.7 represents the geomagnetic activity, TEC, and ΔTEC measurements during the EQ period over an observational period of December 26, 2017, to January 15, 2018. From Fig. 3.7b, we can observe that the selected observational period seemed to be highly affected by geomagnetic activity most of the time. The geomagnetic activities can be observed on December 26, 2017, along with other days such as January

08, 09, 14, and 15, 2018, as the Kp index touches or exceeds the threshold value of 4o. On the other hand, large TEC enhancements were observed on January 1-2, 2018 (9 & 8 days before major shock) as shown by Fig. 3.7c-e, where the geomagnetic activity remains extremely quiet and may be considered as possible pre-EQ seismic anomalies. The values of ΔTEC , which are TEC deviations from UB, at three IGS stations named as MANA, SCUB, and SSIA are recorded as 7.8909, 7.3697, and 7.3811, with a percentage deviation about 53.03%, 55.83%, and 54.36%, respectively, as represented by Table 3.2 and Fig. 3.8. Since the deviations recorded on January 01-02, 2018 (8 days before major shock) have occurred in quiet geomagnetic conditions, so can be considered as seismically induced anomalies. According to Pulinets and Boyarchuk [43], TEC anomalies can be observed up to 12 days before severe and strong EQs.

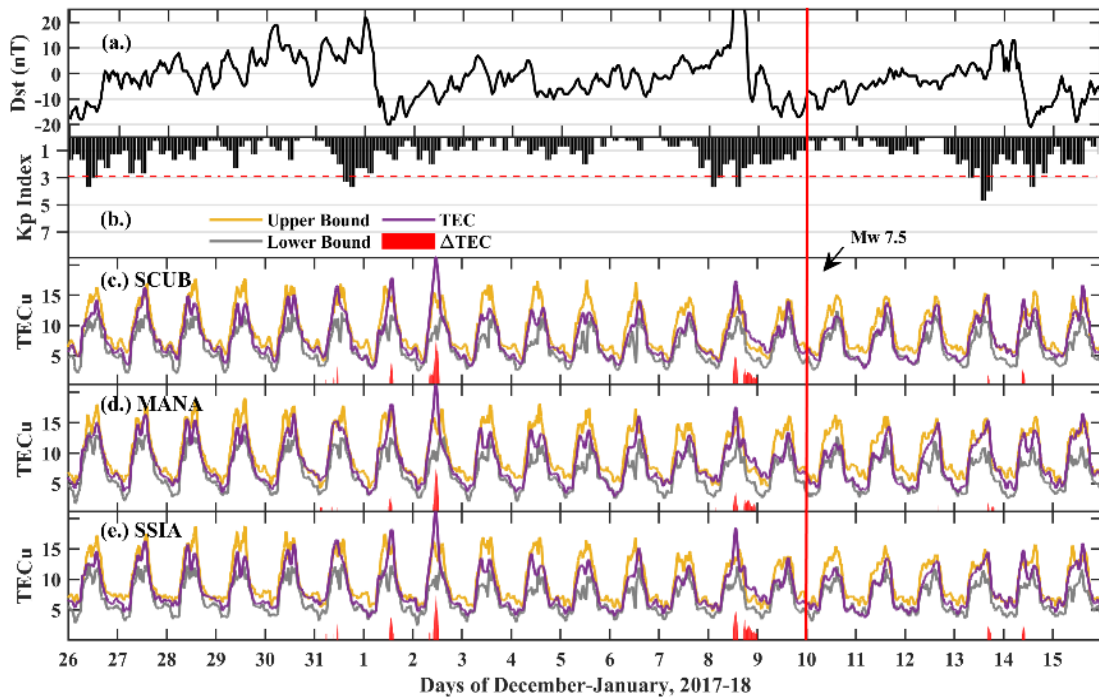


Fig. 3.7. The geomagnetic storm conditions and TEC analysis for three different GNSS stations (panel: c, d, e) for Honduras EQ of M_w 7.5 are showed with detail legend and EQ day.

Table 3.2. Δ TEC anomalous values at different IGS stations used for Greece, Taiwanese, Honduras, and Russia EQs

Event	IGS Station Name	Location of the IGS Station		Δ TEC value	%TEC deviation	Region
M _w		Latitude	Longitude	TECU		
6.8	MATE	40.649°N	16.705°E	2.7546	19.23 %	Italy
	ORID	41.127°N	20.794°E	3.4679	27.61 %	Macedonia
	SOFI	42.556°N	23.395°E	3.7315	32.90 %	Bulgaria
6.1	TWTF	24.954°N	121.165°E	5.6903	32.66 %	Taiwan
	CKSV	22.999°N	120.016°E	7.7575	43.54 %	Taiwan
	KMNM	24.464°N	118.165°E	4.7150	29.56 %	Taiwan
6.4	TWTF	24.954°N	121.165°E	10.2479	52.69 %	Taiwan
	CKSV	22.999°N	120.016°E	8.3275	36.36 %	Taiwan
	KMNM	24.464°N	118.165°E	8.4300	37.64 %	Taiwan
7.5	MANA	12.149°N	86.249°W	7.8909	53.03 %	Nicaragua
	SCUB	59.576°N	75.762°W	7.3697	55.83 %	Cuba
	SSIA	43.529°N	89.116°W	7.3811	54.36 %	El Salvador
7.3	PETS	53.023°N	158.650°E	1.7447	46.26 %	Russia
	MAG0	59.576°N	150.770°E	1.4571	27.53 %	Russia
	STK2	43.529°N	141.846°E	2.7757	43.35 %	Japan
6.2	PETS	53.023°N	158.650°E	1.1710	22.74 %	Russia
	MAG0	59.576°N	150.770°E	1.9099	48.14 %	Russia
	STK2	43.529°N	141.846°E	2.2381	53.48 %	Japan

Moreover, the TEC anomalies were not only observed on January 01-02, 2018 but can also be observed on January 08, 2018, due to some geomagnetic activity. Whereas, Ke et al., [66] claimed in his study, that for large EQs having $M_w \geq 7.0$, extreme geomagnetic conditions may also be the possible precursors to the impending EQs. In contrast, some scientists have refused the EQ forecasting in solar storm conditions as well [52].

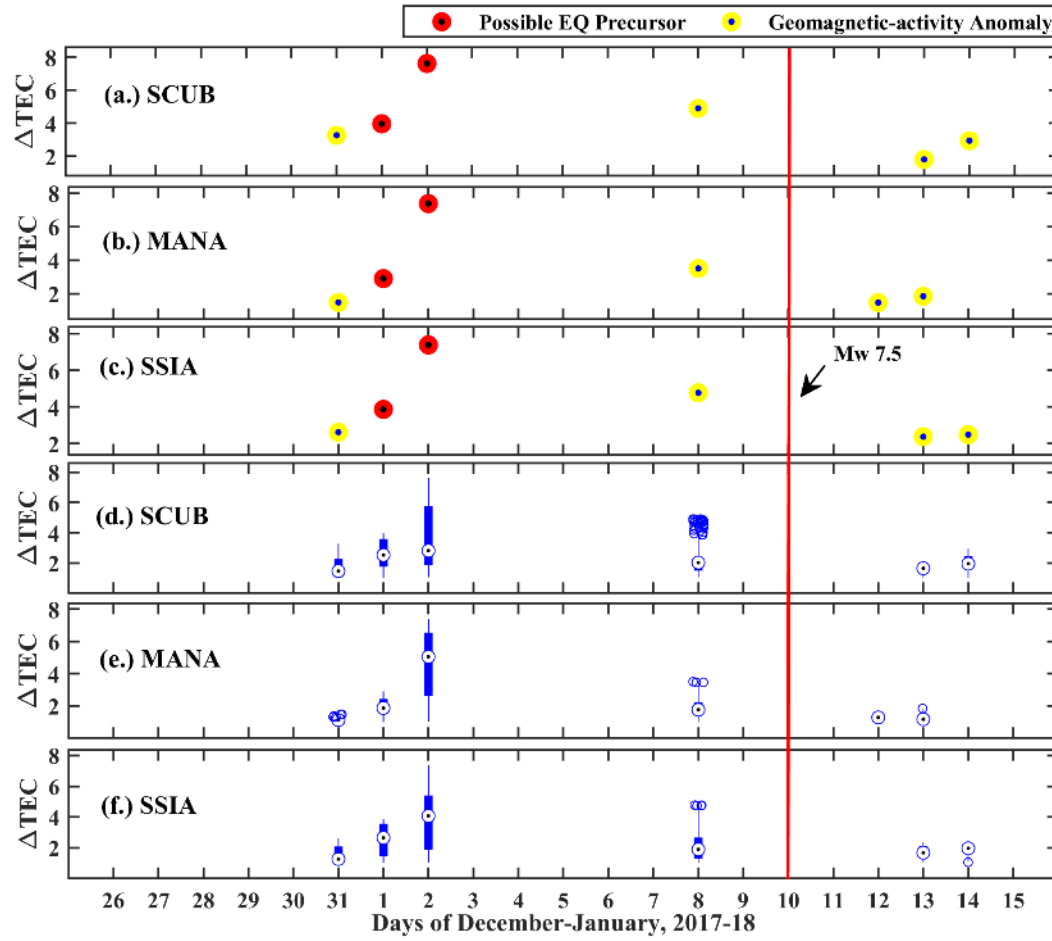


Fig. 3.8. The top three panels show maximum ΔTEC recorded at SCUB, MANA, and SSIA, whereas other panels show a percentage deviation of TEC for Honduras EQs (M_w 7.5).

3.5 TEC Anomalies During the Two Sequential Russian EQs

In the month of December 2018, a large EQ of M_w 7.3 occurred 83 km west of Nikol'skoya, Russia, on December 20, 2018, at 17:01:55 (UTC), having a depth of 16.6 km. This EQ was then followed by an aftershock of M_w 6.2 on December 25, 2018, at 02:10:34 (UTC) with an epicentral depth of 11.2 km. The details of these EQs are given in Table 2.1 and their geographical locations along with the locations of IGS stations used in this analysis are represented by Fig. 2.1. The geomagnetic activities during the EQ period of December 5-30, 2018, and TEC variations along with their associated UB and LB are given in Fig.

3.9. Whereas, Fig. 3.10 depicts the maximum ΔTEC values, differentiating SIAs and geomagnetic activity, at three IGS stations named MAG0, PETS, and STK2. From Fig. 3.9a-b, a geomagnetic activity can be observed on December 28-29, 2018 (8 days after the M_w 7.3 EQ) whose corresponding TEC perturbations can be seen in Fig. 3.9c-e and Fig. 3.10a-c. Apart from these anomalies, other TEC perturbations may also be observed on December 18 and 22, 2018, which are considered as pre-EQ seismic anomalies, as storm activities are found very quiet. Furthermore, Fig. 3.10 scrupulously distinguishes SIAs and GIAs. The maximum ΔTEC values about 1.7447, 1.4571, and 2.7757 TECU were recorded before M_w 7.3 EQ, while maximum ΔTEC values about 1.1710, 1.9099, and 2.2381 were recorded before M_w 6.2 EQ at PETS, MAG0 and STK2 stations, respectively, as shown in Fig. 3,10a-b-c and Table 3.2.

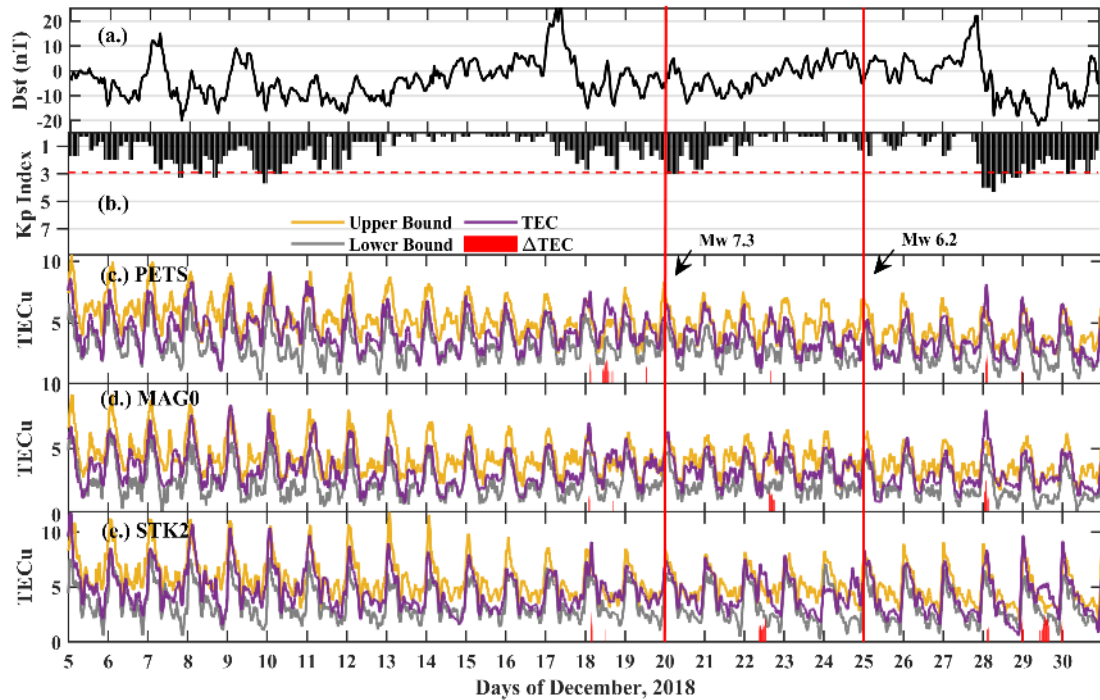


Fig. 3.9. Panel (a and b) shows geomagnetic Dst and Kp Index, panel (c, d, e) shows GNSS TEC along red line for EQs day two sequential Russian EQs. Stations include PETS, MAG0 and STK2.

Similarly, the maximum percentage deviation from UB was recorded as 46.26%, 27.53%, and 43.35% for M_w 7.3 EQ and 22.74%, 48.14%, and 53.48% for M_w 6.2 EQ, at PETS, MAG0, and STK2 stations, respectively. Table 4 further explains the maximum deviations recorded at anomalous days before these EQs. The maximum average ΔTEC values about 1.9925 and 1.7730 TECU with a maximum average percentage deviation about 39.05% and 41.45% are recorded before M_w 7.3 and M_w 6.2 EQs, respectively. Moreover, another exciting phenomenon can also be observed from Fig. 3.10 that, like other sequential EQs studied above, these sequential EQs of Russia also exhibits a proportion to each other as the aftershock also found anomalous exact number of days before its predecessor.

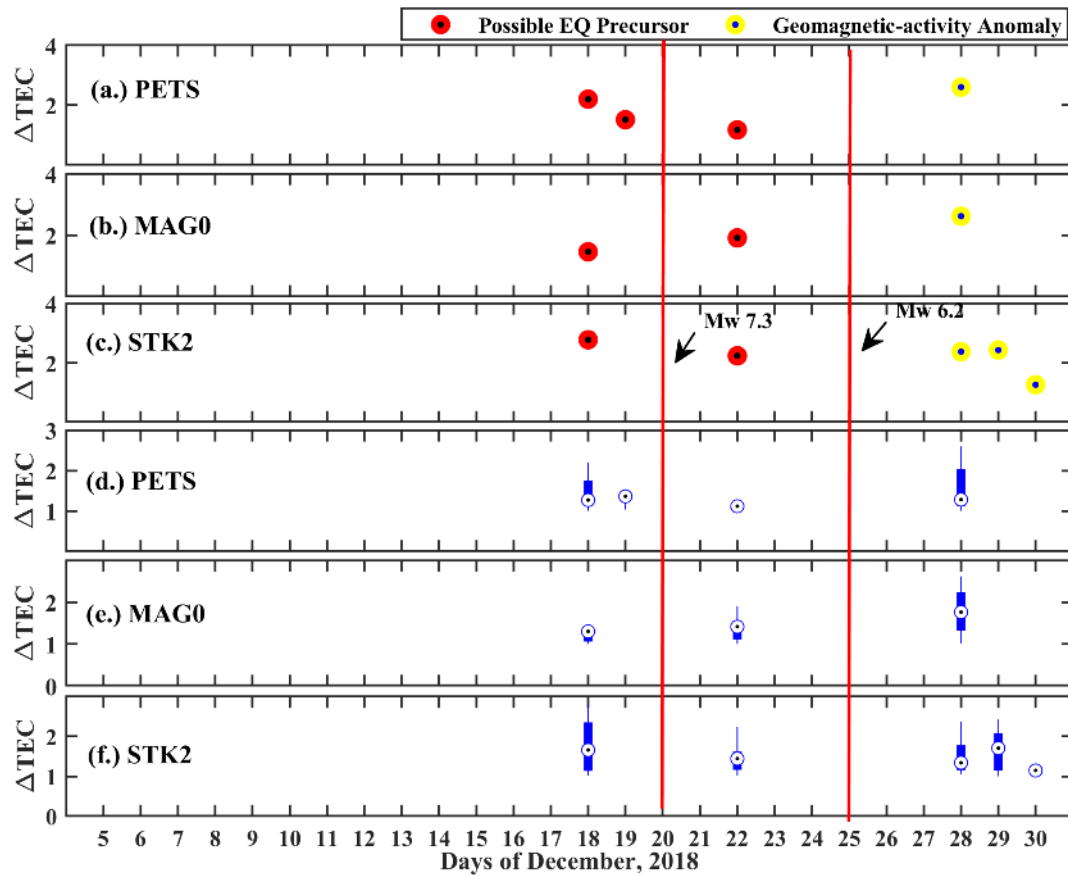


Fig. 3.10. The top three panels show the seismic and storm anomalies with maximum ΔTEC whereas other panels show percentage deviation of TEC for two Russian EQs.

3.6 Ionospheric Responses to the Two Hualien EQs of Taiwan Region

About 22 km north northeast of Hualien, Taiwan, an EQ of M_w 6.1 occurred on February 04, 2018, at 13:56:41 (UTC), having a depth of 12 Km. This EQ was a foreshock of a destructive EQ of M_w 6.4 that occurred on February 06, 2018, about 18 Km north northeast of Hualien, Taiwan, at 15:50:43 (UTC) (Table 2.1, Fig. 2.1f). The ionospheric conditions prior to these back to back EQs are monitored by three IGS stations named as TWTF, KMNM, and CKSV whose geographical locations are given in Table 3.2 and Fig. 2.1f. Moreover, the TEC measurements escorted by their associated UB and LB along with geomagnetic activity during an observational period of Jan 22 – Feb 12, 2018, are represented in Fig. 3.11. From Fig. 3.11c-e, the ionospheric TEC perturbations can be identified on 25 January (10 days before M_w 6.1 EQ), 03 February (1 day before M_w 6.1 EQ) and 05 February 2018 (1 day before M_w 6.4 EQ) at three IGS stations, simultaneously.

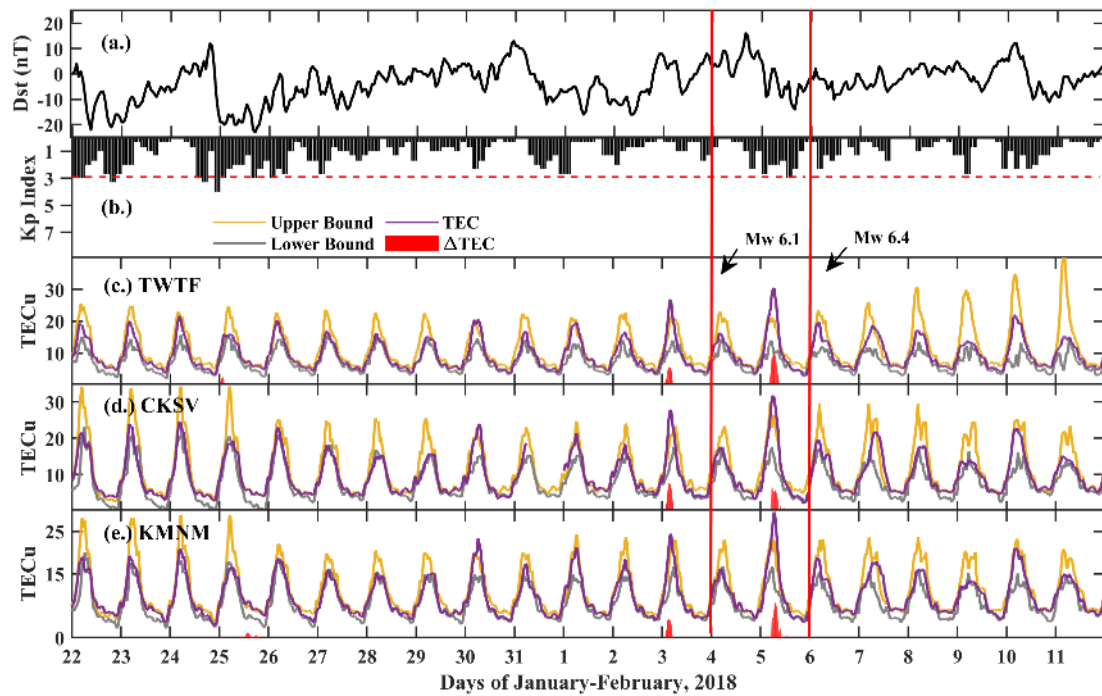


Fig. 3.11. Dst and Kp Index are given in panel (a, b), and TEC measurements from TWTF, CKSV and KMNM stations are showed in context of two sequential Taiwan EQs.

TEC anomalies found on February 03, (1 day before M_w 6.1 EQ) and February 05, 2018, (1 day before M_w 6.4 EQ) are considered as possible pre-EQ seismic anomalies as geomagnetic activity is found quiet during these days, whereas, TEC anomalies found on January 25 (10 days before M_w 6.1 EQ), possibly be the ionospheric response due to abrupt storm origination on this day. The maximum ΔTEC deviations about 5.6903, 7.7575, and 4.7150 TECU with a %TEC deviation about 32.66%, 43.54%, and 29.56% before M_w 6.1, whereas, the maximum ΔTEC deviations about 10.2479, 8.3250, and 8.4300 TECU with a %TEC deviation about 52.69%, 36.36% and 37.64% before M_w 6.4 were recorded at TWTF, CKSV, and KMNM stations, respectively, as shown in Table 3.2 and Fig. 3.12.

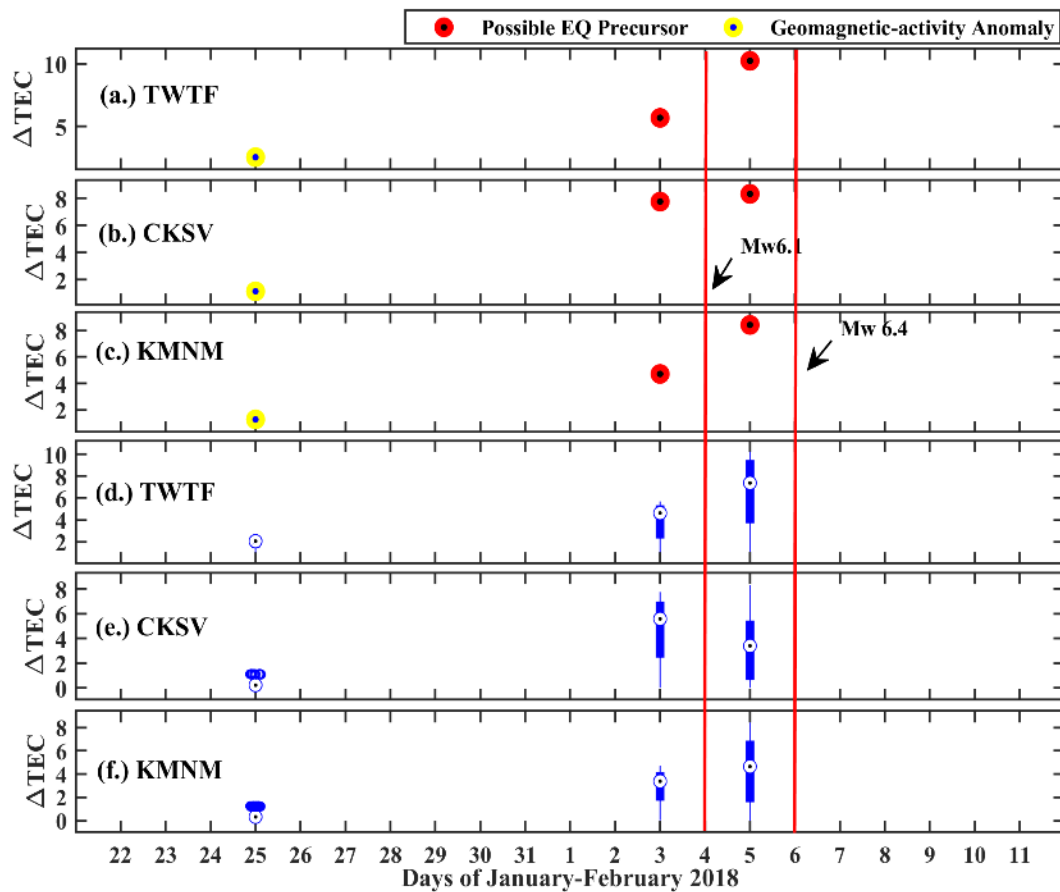


Fig. 3.12. The maximum ΔTEC and percentage deviation of GNSS-TEC measurements for the two sequential Taiwan EQs are showed.

Liu et al. [67] statistically investigated pre-EQ anomalies around Taiwan that includes 184 EQs having $M_w \geq 5.0$ during 1994-1997 and found negative anomalies within 5 days before the impending EQs. Similarly, On the other hand, in our analysis, ionospheric conditions have found variant 1 day before impending EQs as shown in Fig. 3.12. These Taiwan EQs of 2018, have also shown synchronized pattern of anomaly exhibition as both back to back EQs exhibited anomalous ionosphere exactly 1 day before their main shocks. Similar findings were reported by Nishihashi et al. [68], they manifested the same kind of ionospheric variations where they find anomalous ionospheric TEC 1 and 3 days before M_w 6.4 Chia-Yi EQ in Taiwan.

3.7 Discussion and Summary

In this thesis, the possible SIAs before 11 EQs, that have occurred in 2018, have been investigated using GNSS-TEC time-series from a total of 18 IGS stations. Many studies in past have also reported the occurrence of TEC anomalies within 5-10 days during the EQ preparation period using some statistical techniques [15, 16, 44, 48, 50, 65]. For example, the SIAs have been statistically analyzed before several large and severe EQs that can be observed up to 12 days before the impending EQs [43]. Similarly, the focal mechanism behind the generation of these anomalies have been discussed briefly in the previous sections. Like, Kuo et al. [58] simulated the ionospheric electron density for the coupling of the lithosphere and ionosphere by numerical procedures and the EQ conductivity was driven to ionosphere by an upward electric current. Furthermore, they portrayed the increase/decrease in ionospheric perturbations due to ions emanation over the epicenter due to the plasma flow direction. These findings urged us to investigate the ionospheric conditions during quiet geomagnetic activity before EQs.

This investigation of the ionosphere before the EQs is two-fold: first to explore the SIAs and second is to discriminate these anomalies with geomagnetic activities during the analysis. Table 3.3 provides a brief overview of this differentiation. Similarly, Fig. 3.13 delineates the final statistics about the TEC deviations in terms of seismically and GIAs. For Example, in case of three South American EQs, GIAs showed an average deviation about 26.68% from UB, whereas SIAs show an average value about 33.54%, 36.41%, and 33.13% before M_w 5.4, M_w 5.2 and M_w 5.6 EQs, respectively.

Table 3.3. Overall Percentage Δ TEC deviations for seismic ($K_p \leq 3$) and storm ($K_p > 3$) anomalies

Event	Ave. Max. Δ TEC of SIA	Ave. Max. % Δ TEC of SIA	Ave. Max. Δ TEC of GIA	Ave. Max. % Δ TEC of GIA	Max. Value of K_p Index
5.6	1.7193	59.62 %	2.7431	88.35 %	4-
7.1	1.3756	52.20 %	2.0459	69.35 %	4-
5.4	6.5606	33.54 %	5.6546	26.68 %	4o
5.2	5.4180	36.41 %	5.6546	26.68 %	4o
5.6	6.1690	33.13 %	5.6546	26.68 %	4o
6.8	3.3180	26.58 %	N/A	N/A	4+
7.5	7.5472	54.41 %	4.3924	38.96 %	4-
7.3	1.9925	39.05 %	1.1133	29.18 %	4-
6.2	1.7730	41.45 %	1.1133	29.18 %	4-
6.1	6.0543	42.75 %	2.7532	19.82 %	4o
6.4	9.0018	45.69 %	2.7532	19.82 %	4o

The Greece EQ exhibits an average deviation about 3.3180 TECU and a percentage deviation of 26.58% from UB, though any GIA was not observed. Similarly, in the case of M_w 7.5 EQ, a maximum average ΔTEC value for SIA was recorded as 7.5472 with maximum average percentage deviation of 54.41%, whereas, on January 08, 2018, a maximum average ΔTEC value 4.3924 with maximum average percentage deviation of 38.94% was recorded. So, it evident that the seismically induced anomalies are more deviated than geomagnetically induced anomalies. Furthermore, we can observe that on December 09, 2018, which is considered as geomagnetically disturbed day, the maximum average ΔTEC value about 1.1133 with a maximum average percentage deviation of about 29.18% is recorded during the observational period of the Russian EQs. Whereas, on December 18 and 23, 2018, which are the possible seismic precursors to the EQs, the maximum average ΔTEC value about 1.9925 and 1.7730 TECU with a maximum average percentage deviation about 39.05% and 41.45% is recorded before M_w 7.3 and M_w 6.2 EQs, respectively. Considering the Taiwanese twin EQs, an average maximum ΔTEC value of 6.0543 and 9.0018 TECU with a %deviation about 42.75% and 45.69% was recorded on anomalous days before M_w 6.1 and M_w 6.4 EQs, respectively, as shown in Table 3.3. Additionally, Table 3.3 also highlights the fact that average maximum ΔTEC deviation recorded on January 24, 2018, due to a geomagnetic activity was about 2.7532 TECU with a %deviation of 19.82% and deviated less than SIAs.

On the other hand, the situation is a bit different for the Alaskan EQs than others. For Alaska EQs, the average maximum ΔTEC values of SIAs for M_w 5.6 and 7.1 EQs were recorded as 1.7193 and 1.3756, respectively, with a percentage of 59.62% and 52.20% from the corresponding UB. Similarly, for GIAs, the average maximum values and

percentages are 2.7431 and 88.35% during M_w 5.6 EQ, whereas 2.0459 and 69.35 % are during M_w 7.1 EQ, respectively. The reason behind the significant GIAs are the location of the IGS stations. Alaska state is located within the polar regions where the geomagnetic effects are more dominant than equatorial latitudes. So, one may expect the higher deviations in the TEC even due to a low geomagnetic activity. From Fig. 3.13, it can be observed that possible seismic anomalies have exhibited a TEC deviation of about 4.6299 TECU from UB, averagely. Whereas, the geomagnetically induced anomalies have shown a relatively less deviation of about 3.3878 with a percentage deviation of 37.47% averagely. However, it should be noted that these deviations are only induced by moderate geomagnetic activity ($K_p=4$), but, higher geomagnetic activities will, of course, induce more intense perturbations.

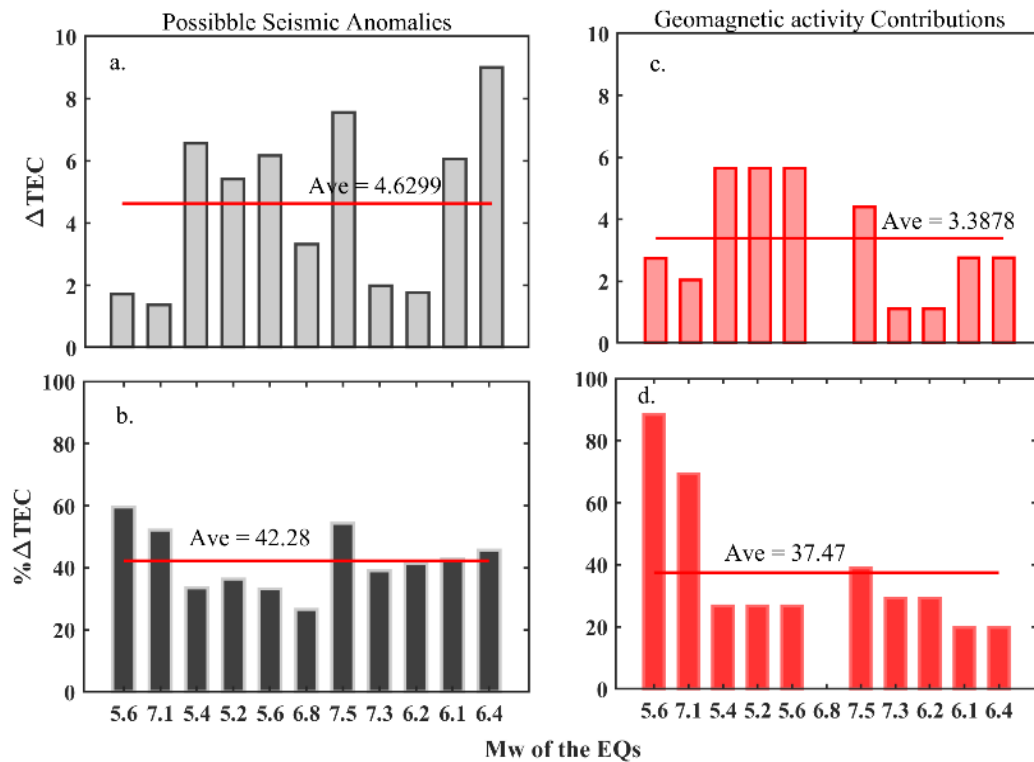


Fig. 3.13. Deviation in TEC for seismic anomalies and storm anomalies in top panel and percentage deviations from UB bounds in lower panel for all EQs studied in this study.

Furthermore, some other reports classified the thermal heating of the ionosphere from the lower atmosphere during quiet geomagnetic activities and showed that seismic anomalies normally do not deviate more than 20% [69, 70]. In contrast, our results and Hasbi et al. [71] showed a maximum deviation of more than 20 % (see Fig. 3.13) as EQ enhancement. Another interesting and unique phenomenon associated with sequential EQs can be observed in Fig. 3.14. It can be observed that the sequential EQs, including fore and aftershocks, exhibit ionospheric anomalies at the almost same number of days as its predecessor. On the other hand, some reports strongly rejected the existence of EQ ionosphere anomalies with GNSS TEC [52, 53]. However, more studies accepted the execution of EQ induce ionosphere anomalies with GNSS TEC. For example, Oikonomou et al. [72] analyzed ionospheric anomalies during the EQ preparation period of M_w 8.3 Chile EQ and found them within 1-5 days before the mainshock. All these reports emphasized on more analysis of TEC monitoring during $K_p < 3$ storm conditions with a well-equipped cluster of the ground and satellite-based observations for EQ precursors.

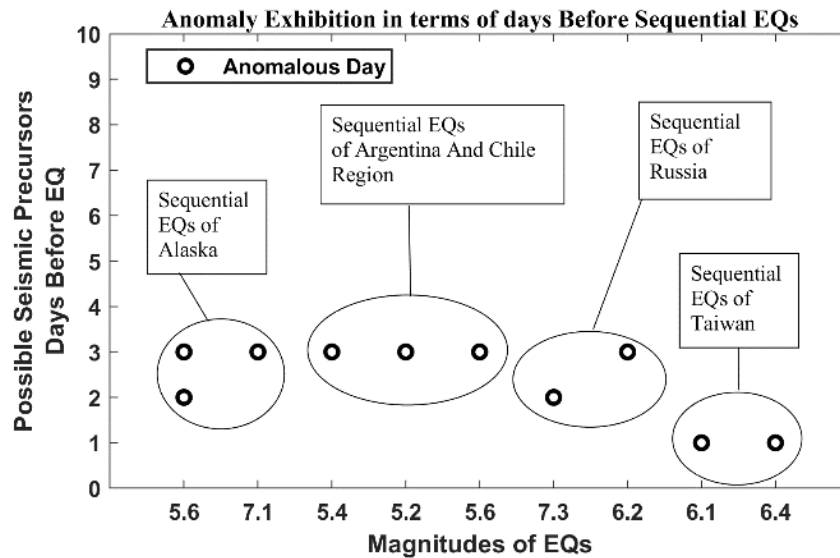


Fig. 3.14. The anomaly pattern followed by sequential EQs. EQs sharing same regions exhibit strong proportion in term of anomaly exhibition and follows each other.

4. Conclusion and Future Aspects

We study the possible relation of EQs and ionospheric anomalies from GNSS TEC of global $M_w > 5.0$ EQs during the periods of a low geomagnetic activities of less than 3 Kp in 2018. Results suggest that abrupt seismo TEC variations during $K_p \leq 3$ geomagnetic activities appeared as a result of the emanation of energy from the EQ prone region to the ionosphere. We also find evidence of some moderate geomagnetic activities ($K_p > 3$) induced anomalies in the analysis of the TEC time series. The key findings from our analysis are:

- Positive TEC anomalies occur within 5 days before and after all $M_w > 5.0$ except $M_w 7.5$, Honduras EQ from TEC measurements of GNSS station within the preparation zone. However, some ionospheric anomalies tend to proceed till 10 days before the mainshock due to abnormal emanation of energy, like in the case of $M_w 7.5$, Honduras EQ.
- We find more than 4 TECU deviation of daily TEC beyond UB for seismo ionospheric anomalies and about 3.7878 for moderate geomagnetic activities ($K_p=4$) induced anomalies. This corresponds to 42.28% for EQs anomalies and 37.47% for the geomagnetic activity in percentage deviation within the analysis period of GNSS TEC.
- Moreover, a strong tendency of EQ ionosphere variations is recorded than moderate geomagnetic variations during the analyzed period before and after the mainshocks. These results suggested that seismo ionosphere anomalies may exist but the need

for more dedicated ground and space observations will further clarify the phenomenon.

Our results provide some evidence about the existence of the SIAs before all investigated EQs. However, we emphasize on more analysis of TEC monitoring during $K_p < 3$ storm conditions with a well-equipped cluster of the ground and satellite-based observations for the search of ionospheric EQ precursors. These days, large clusters of dual-frequency GNSS receivers are developed within the countries called Continuously Operating Reference Stations (CORS). Using CORS networks, one may investigate the ionospheric responses at more detailed and broader platform by collecting long-term timeseries. Similarly, considering the future works, a large network of CORS may also help in monitoring the slip displacement during a seismic event. Simultaneous observations of the InSAR and GNSS may also be analyzed for better understanding of the seismic processes. Furthermore, some frequency wavelet and spectral analysis of GNSS-TEC may also be conducted to understand the deep morphology of the electron concentrations.

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